Statistical Studies in Formula 1

Lap-Time Distributions, Grid-Position Advantage, Home-Race Effects, and Pit-Stop Strategy (1950–2026)

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Abstract

We apply classical inference and modern machine learning to the publicly available Ergast and FastF1 Formula 1 datasets, addressing four self-contained research questions. *First*, we ask which parametric distribution describes within-race lap times: across circuits and regulation eras, right-skewed families (Gamma, Log-Normal) consistently outperform the Normal under AIC and BIC criteria. *Second*, we measure the predictive power of qualifying position: pole converts to victory 43% of the time, and each grid slot lost costs a driver, on average, 0.45 finishing positions. *Third*, we test for a home-crowd advantage using a teammate-relative target that removes the dominant constructor confound; a paired t-test finds no aggregate effect, but a mixed-effects model uncovers a small but significant home$\times$experience interaction. *Fourth*, we predict pit stops from in-race telemetry with a bidirectional LSTM, characterise the pit hazard with a Cox proportional-hazards model, and classify overtakes as strategic or on-track. Across the four studies a single methodological message emerges: in a sport dominated by machinery and regulation drift, the choice of *target variable* matters at least as much as the choice of model.

Contents

Headline numbers. A one-page summary of the four studies appears in Table 1.

Table 1. Headline results across the four studies.

Study	Target	Headline result	Sample
Lap-time distributions	Within-race lap time	Right-skewed families dominate; Gamma minimises AIC on a typical race (1921.3 vs. 1933.6 for Normal)	618,766 laps, 571 races
Starting-grid advantage	Finishing position	$P(\text{Win} \mid \text{Pole}) = 43.0\%$; OLS slope $\hat{\beta}_{\text{grid}} = 0.45$ per slot	25,646 driver–races
Home-race advantage	Era-adjusted teammate delta	No aggregate effect ($t = 0.65$, $p = 0.51$); significant home$\times$experience interaction	27,304 driver–races
Pit-stop strategy	Binary pit on next lap	BiLSTM assigns mean prob. 83.8% on true pit laps; 3.16% positive rate	152,475 laps, 2018–2025

1 Introduction

Formula 1 generates a uniquely rich record: more than seventy years of race results, lap-by-lap telemetry from the modern era, and a stable set of identifiers that link drivers, constructors, and circuits across the decades. The sport is also unusually difficult to model. Three features recur in every analysis below.

1. **Machinery dominates.** Roughly 70–80% of the variance in finishing position is attributable to the constructor. Any naive regression of outcome on driver-level covariates is swamped by this single confounder.
2. **Regulations drift.** Engine formulas, aerodynamic packages, tyre suppliers and the points system have all changed many times. Pooling raw points across eras is statistically incoherent.
3. **Outcomes are discrete and bounded.** A driver can finish in at most twenty-something positions, and the marginal importance of one position depends on where on the grid we are. A loss to standard linear assumptions follows.

Each study below addresses one of the standard inferential or predictive questions about F1, and each adopts a target variable designed to be robust to (1)–(3). Section 2 fits parametric distributions to within-race lap times. Section 3 quantifies the advantage of starting near the front. Section 4 isolates the home-race effect via an era-adjusted teammate delta. Section 5 builds a deep-learning pit-stop predictor and embeds it in a wider survival-analytic framework. Section 7 synthesises the findings. The appendix documents data sources, software, and reproducibility.

Data overview. The shared foundation is the Ergast database (Kaggle redistribution), containing 14 relational CSVs spanning 1950–2024 with the 2025–2026 schedule. Standard inner joins of `results`, `races`, `circuits`, `drivers`, and `status` yield 27,304 driver–race rows used by Sections 3 and 4. Section 2 additionally exploits the `lap_times` table, which records 618,766 individual laps across 571 races since 1996. Section 5 augments the historical record with full per-lap telemetry from FastF1 (2018–2025).

Notation. Throughout, i indexes drivers, r indexes races, and c indexes circuits. The shared statistical notation is collected in Table 2; section-specific notation is introduced locally.

Table 2. Common notation used throughout the report.

Symbol	Meaning
i, r, c	Driver, race, and circuit indices
$p_{i,r}$	Points scored by driver i in race r
$p^{\star}_{i,r}$	Era-normalised points $p_{i,r} / \max_j p_{j,r} \in [0, 1]$
$\Delta_{i,r}$	Teammate delta (Section 4)
$H_{i,r}, E_{i,r}$	Home-race indicator and experience years
$\hat{\beta}, \hat{\theta}$	Estimated regression/parametric coefficients
σ_u^2, σ^2	Random-intercept and residual variance
AIC, BIC	Information criteria, $2k - 2\ell$ and $k \ln n - 2\ell$

2 Lap-Time Distributions

2.1 Question and motivation

Many practical tasks in motorsport — spotting laps that were unusually slow or fast, simulating race strategy, comparing drivers fairly — need a statistical model of a single lap time. The default model is the Normal (bell-curve) distribution, but lap times don't behave symmetrically. There is a hard floor on how fast a car can lap a given track, set by the limits of the engine, the tyres, and the track itself; but there is no ceiling on how slow a lap can be. A driver stuck behind a slower car, a section of track taken cautiously, worn tyres, or a small on-track mistake can all add seconds to a lap. We therefore expect a *right-skewed* distribution — one with a longer tail on the slow side — rather than a symmetric bell curve.

We ask three concrete questions:

1. Which of three standard distributions (Normal, Log-Normal, Gamma) best describes the lap times within a single race?
2. Does the answer depend on the type of circuit?
3. Does it depend on the regulation era (the set of technical rules in force at the time)?

2.2 Data and cleaning

For each race we use the per-driver per-lap times from the `lap_times` table. Two filters are applied before fitting.

- We drop lap 1, the first lap of the race. Unlike the subsequent laps, which are run continuously at racing speed, lap 1 begins from a complete standstill on the starting grid: the drivers have to accelerate from rest, and lap 1 also has a much higher rate of collisions and other on-track incidents than later laps, because all the cars are still packed close together at the start, before they have had time to spread out along the track. Both effects make lap 1 systematically slower and more variable than a normal lap, so including it would distort the distribution.
- We also trim laps affected by anything other than normal racing. Drivers entering the pits (where the team changes tyres or repairs damage) slow down well before they reach the pit-lane entry, so the lap on which they pit and the lap on which they rejoin the track are

both several seconds slower than a normal racing lap. The same is true of laps run behind the *safety car* — an official car that is sent onto the track at greatly reduced speed after a serious incident, with all the competing cars required to follow it in single file until the track is clear. The dataset does not flag these laps directly, so we remove them statistically using the interquartile-range (IQR) rule: we discard any lap whose time falls outside

$$[\,Q_1 - 2\cdot\mathrm{IQR},\quad Q_3 + 2\cdot\mathrm{IQR}\,],$$

where Q_1 and Q_3 are the 25th and 75th percentiles of the lap times in the race and $\mathrm{IQR} = Q_3 - Q_1$ is the gap between them. We use a multiplier of 2 rather than the textbook 1.5 deliberately: the tighter 1.5 cutoff would also remove genuinely slow racing laps, such as laps run on heavily worn tyres at the end of a tyre stint (the stretch of laps a driver completes on a single set of tyres), which we want to keep.

For the 2023 Italian Grand Prix at Monza, this removes 56 of the 936 raw laps (6.0%), leaving 880 clean laps for fitting.

2.3 Candidate distributions

We try three standard distributions, each capturing a different shape.

- **Normal.** The familiar bell curve, symmetric around its mean. This is the textbook default for noisy measurements.
- **Log-Normal.** A right-skewed distribution: most values cluster near the low end, with a long tail of larger values. It can only take positive values, which suits a quantity like lap time (which cannot be zero or negative).
- **Gamma.** Also right-skewed and positive-only, but with a lighter upper tail than the Log-Normal — it allows for asymmetry, but does not permit laps that are very far from the mean. A common choice in engineering for strictly positive quantities such as waiting times.

Both Log-Normal and Gamma can capture the asymmetry we expect from lap times; the Normal cannot.

2.4 How we judge a fit

For each candidate distribution we ask: how well does it match the data we observed? We rank the three candidates using two standard scores, **AIC** (Akaike Information Criterion) and **BIC** (Bayesian Information Criterion). Both reward closeness to the data while penalising a distribution that uses more parameters — so a more complicated distribution has to fit noticeably better to win. Smaller values are better, and the distribution with the smallest AIC (and BIC) is declared the winner. We do not also report formal goodness-of-fit tests: the histogram and CDF figures below already show, visually, whether the winning distribution actually hugs the data well.

2.5 Deep dive: 2023 Italian Grand Prix

Figure 1 overlays the three fitted distributions on the histogram of cleaned lap times from the 2023 Italian Grand Prix. The Log-Normal and Gamma curves are nearly indistinguishable through the *body* of the distribution (the middle range, where most laps lie); they only separate in the *tails* (the extreme fast and slow laps) and on the goodness-of-fit scores.

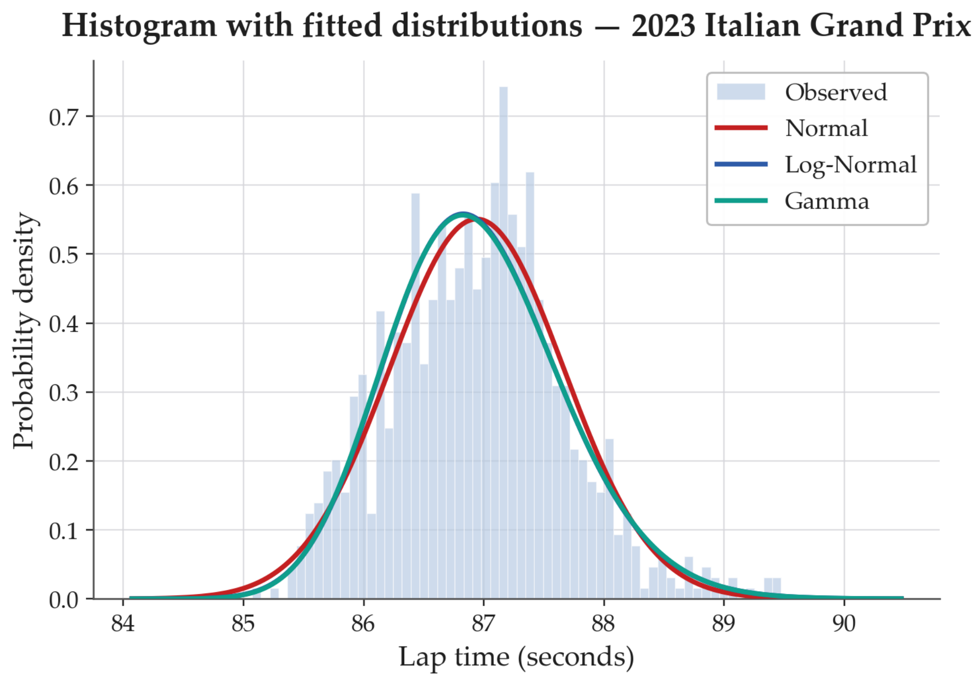


Figure 1. Empirical histogram and three fitted PDFs for the 2023 Italian Grand Prix ($n = 880$).

Figure 2 shows the AIC and BIC ranking. The Gamma fit minimises both information criteria, with Log-Normal a near-tie.

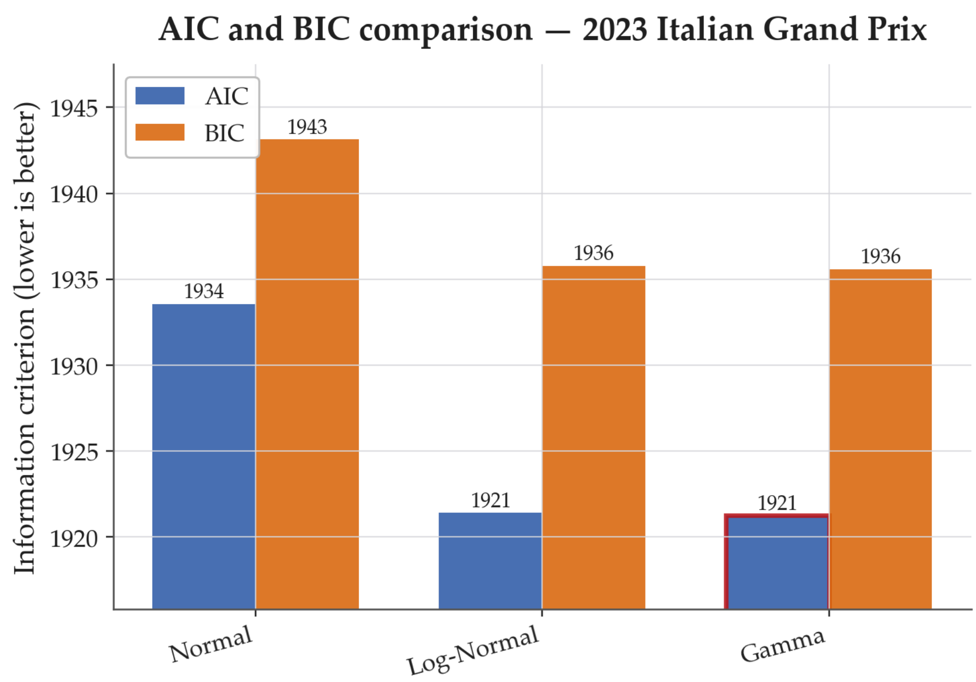


Figure 2. AIC and BIC comparison across the three candidate distributions. The Gamma fit (red outline) wins on both criteria.

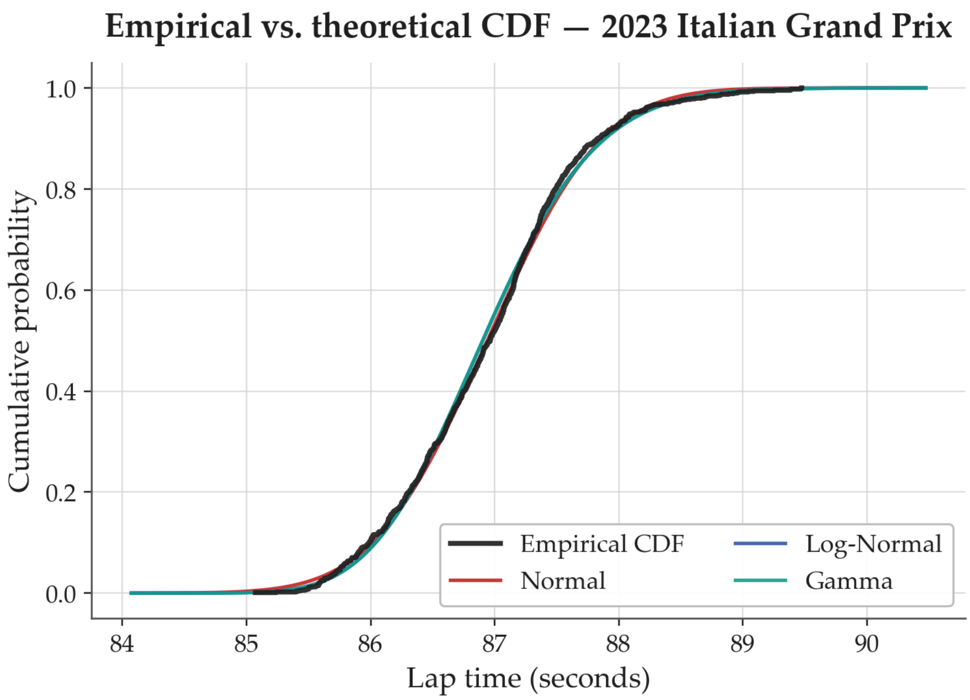


Figure 3. Empirical (data) vs. theoretical (fitted) cumulative distribution functions, or CDFs. A CDF gives, for each lap time x, the fraction of laps that are no slower than x; it rises from 0 to 1. The black step function is the empirical CDF computed directly from the data. The right-skewed families (Gamma, Log-Normal) hug it more closely than the Normal.

Table 3 reports the AIC and BIC scores for the three candidates. The Gamma distribution attains the smallest AIC and BIC, with Log-Normal a near-tie; the Normal is third.

Table 3. AIC and BIC scores, 2023 Italian Grand Prix. Smaller values indicate a better fit.

Distribution	AIC	BIC
Gamma	1921.3	1935.6
Log-Normal	1921.4	1935.8
Normal	1933.6	1943.1

The size of the gap between Normal and the two right-skewed families ($\approx$ 12 AIC points) is substantial: it tells us the asymmetry is a real feature of the data, not just noise. The Gamma–Log-Normal gap (0.1 AIC points), by contrast, is negligible — on this race we cannot tell the two apart.

2.6 Cross-circuit comparison

We repeat the procedure on two circuits from the same season, chosen to span very different circuit types:

- **Monza** — a long, fast, purpose-built racing circuit with long straights and only a few corners.
- **Monaco** — a slow, narrow circuit laid out on closed-off public streets through the city of Monte Carlo. The track is lined by concrete walls and there is almost no room to overtake.

Figures 4 and 5 show each circuit's lap-time distribution and best-fit curve.

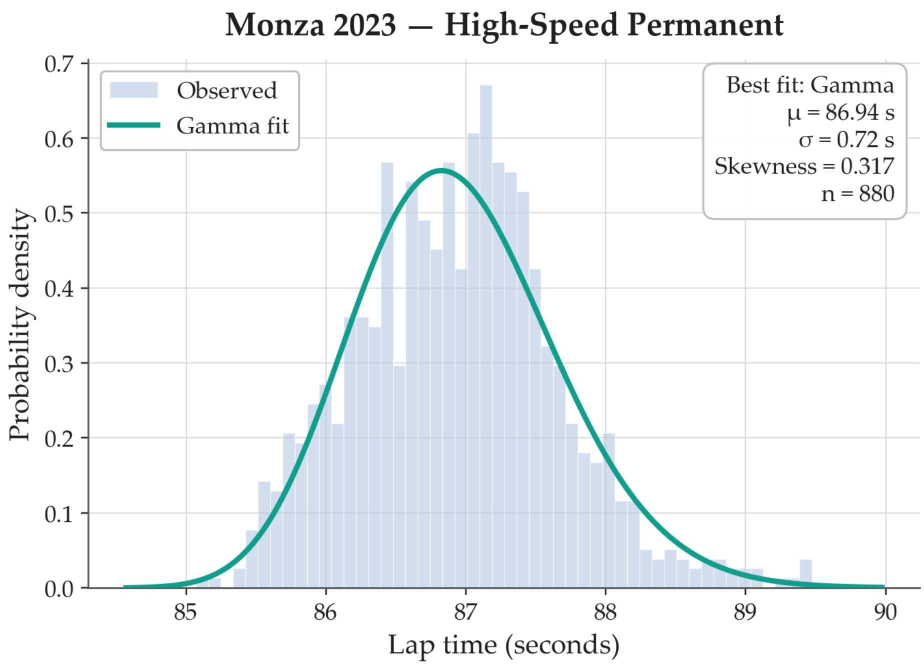


Figure 4. 2023 Monza — purpose-built, high-speed circuit. Lap times are tightly clustered and only mildly asymmetric; Gamma is the best fit.

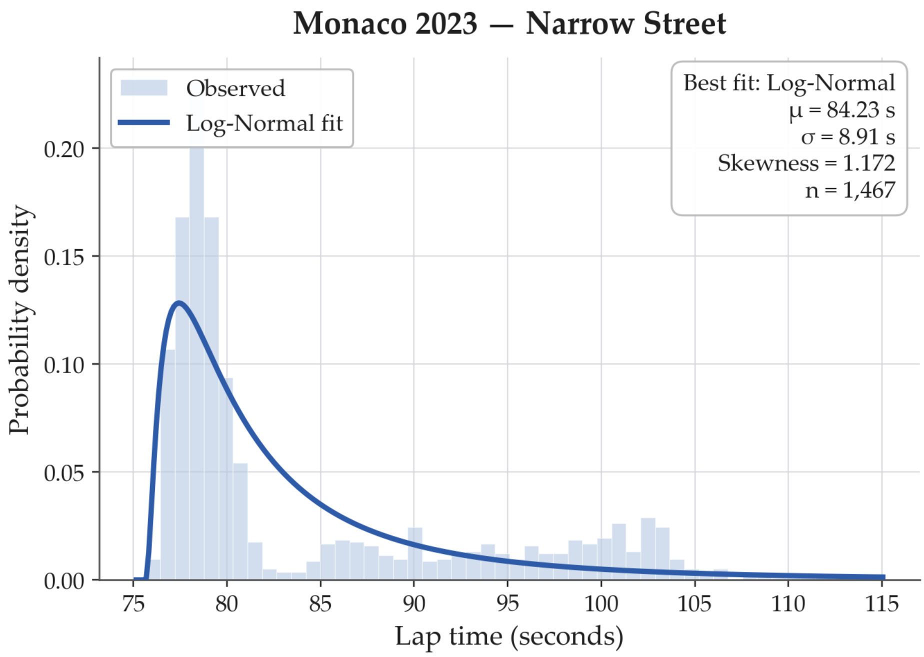


Figure 5. 2023 Monaco — narrow street circuit. Lap times are spread out roughly ten times more than at Monza, with a long slow tail produced by traffic in the cramped layout; Log-Normal is the best fit.

A summary table of the best fit per circuit appears in Table 4.

Table 4. Best-fit summary by circuit, 2023 season.

Circuit	n	μ (s)	σ (s)	skew	Best fit (AIC)
Monza	880	86.94	0.72	0.32	Gamma
Monaco	1,467	84.23	8.91	1.17	Log-Normal

The contrast is striking. (In the table, μ is the mean lap time, σ is the standard deviation — a measure of how much the lap times typically vary from the mean — and "skew" is a measure of asymmetry that is 0 for a symmetric distribution and positive when the slow tail is the longer one.) Monaco's lap times are roughly ten times more spread out than Monza's ($\sigma = 8.91$ s vs. 0.72 s) and visibly more asymmetric (skew 1.17 vs. 0.32). The Log-Normal fit at Monaco reflects the long slow tail produced by drivers caught behind a slower car on a track that has almost nowhere to overtake; the milder Gamma fit at Monza reflects a track where overtaking is easier and the slow tail is shorter. The choice of best-fit family therefore depends on the circuit, not just on the cars or drivers.

2.7 Cross-era comparison at Monza

Keeping the circuit fixed at Monza and varying the year lets us isolate the effect of the technical regulations: any change we see between years is due to the rules and the cars they produced, not to a change of venue. Figures 6–9 compare four Italian Grands Prix run under four different sets of rules.

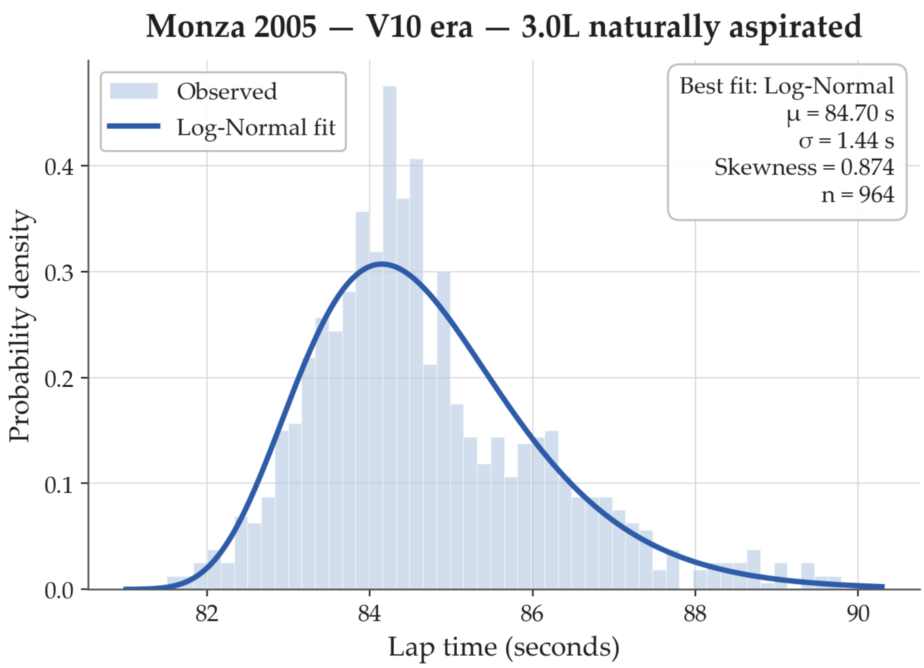


Figure 6. Monza 2005 — V10 era. Wide distribution and high skewness; the AIC selects Log-Normal.

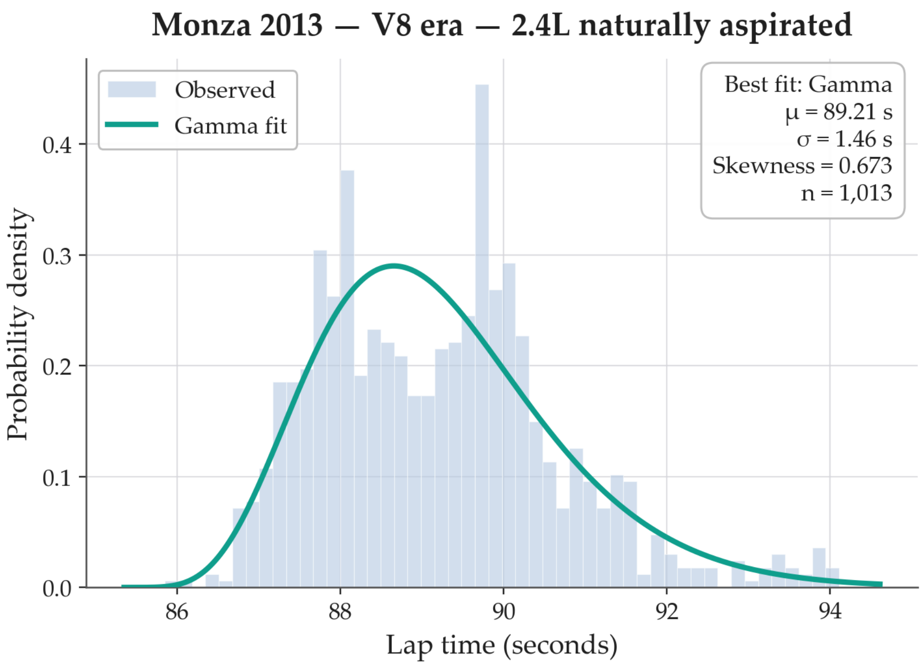


Figure 7. Monza 2013 — V8 era. Slower in absolute terms but similar dispersion; Gamma is AIC-optimal.

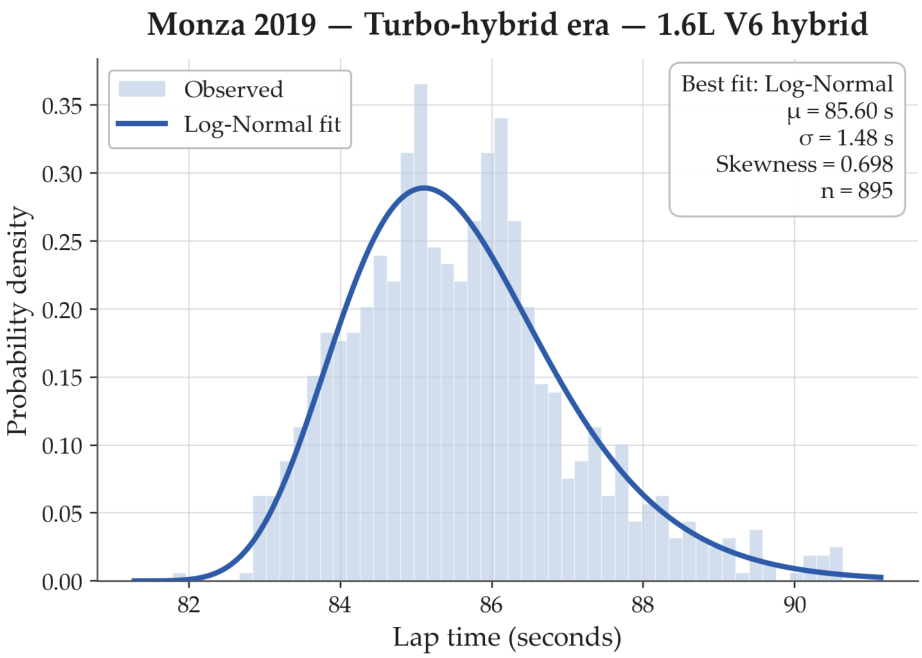


Figure 8. Monza 2019 — turbo-hybrid era. Log-Normal best fit.

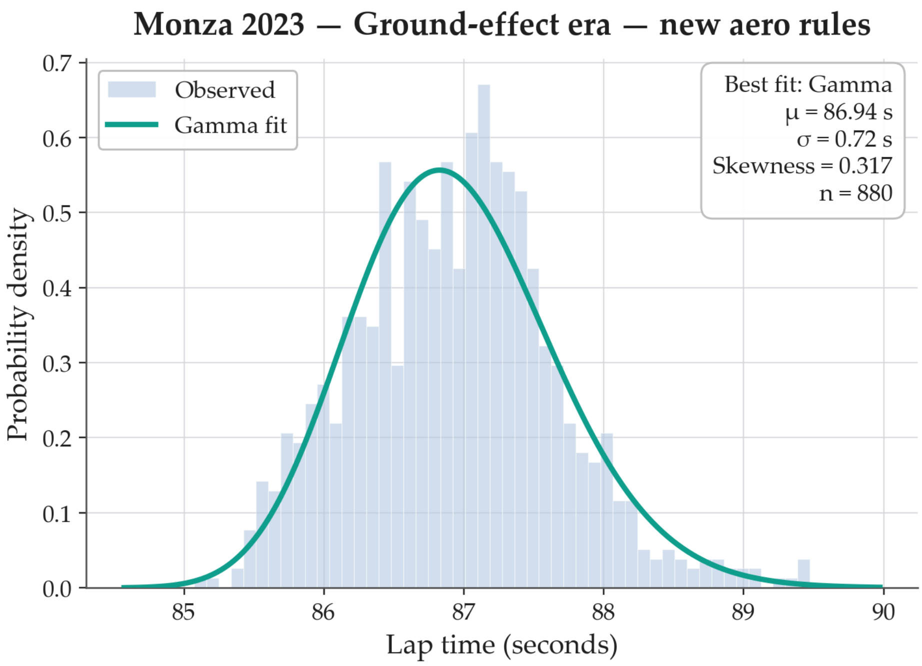


Figure 9. Monza 2023 — ground-effect era. Visibly tighter than earlier eras; Gamma best fit.

The absolute pace shifts with the rules — the V10 cars of 2005 lap roughly two seconds faster than the V8 cars of 2013, and the turbo-hybrid and ground-effect cars sit in between — but the *shape* of the distribution is stable. Table 5 collects the summary statistics. Every era remains right-skewed: even in the modern ground-effect era the distribution is best fit by Gamma, not Normal. The modern era is, however, noticeably tighter (the standard deviation is 0.72 s vs. 1.44 s in the V10 era). This is consistent with two changes in modern racing: car designs across teams have become much more similar, and the narrow temperature range over which modern tyres grip the track well punishes any driver whose tyres fall outside it.

Table 5. Best-fit summary by era at Monza.

Year	Era	n	μ (s)	σ (s)	skew	Best fit
2005	V10	964	84.70	1.44	0.87	Log-Normal
2013	V8	1,013	89.21	1.46	0.67	Gamma
2019	Turbo-hybrid	895	85.60	1.48	0.70	Log-Normal
2023	Ground effect	880	86.94	0.72	0.32	Gamma

2.8 Synthesis: why right-skew is the rule

Figure 10 ranks the skewness across all seven races analysed; Figure 11 shows which family wins by AIC most often. Every race has positive skew, and only one of the seven (Silverstone) is best fit by the Normal.

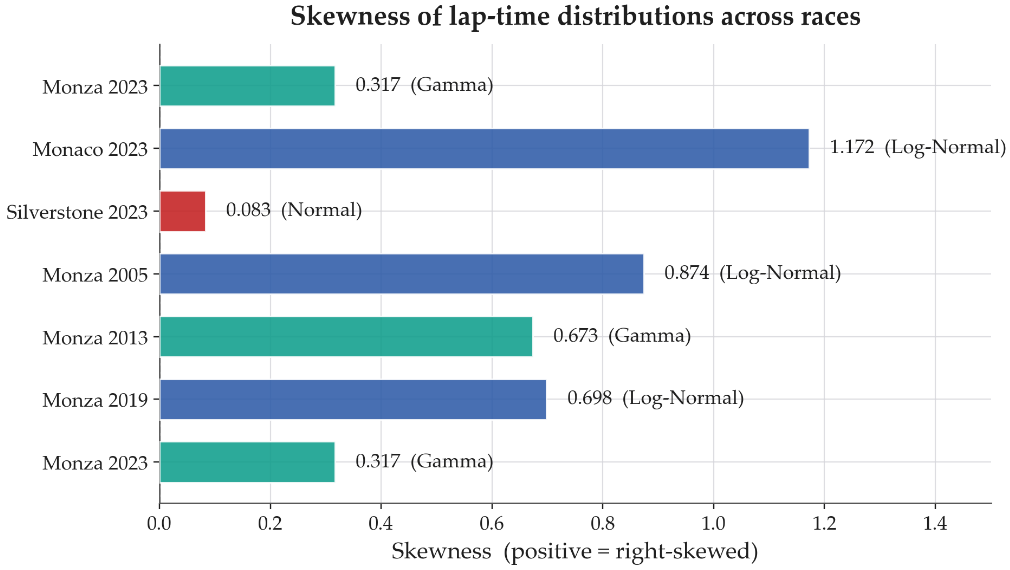


Figure 10. Skewness of every race analysed, bar coloured by AIC-best family. All skews are positive; Monaco is by far the most skewed, Silverstone the least.

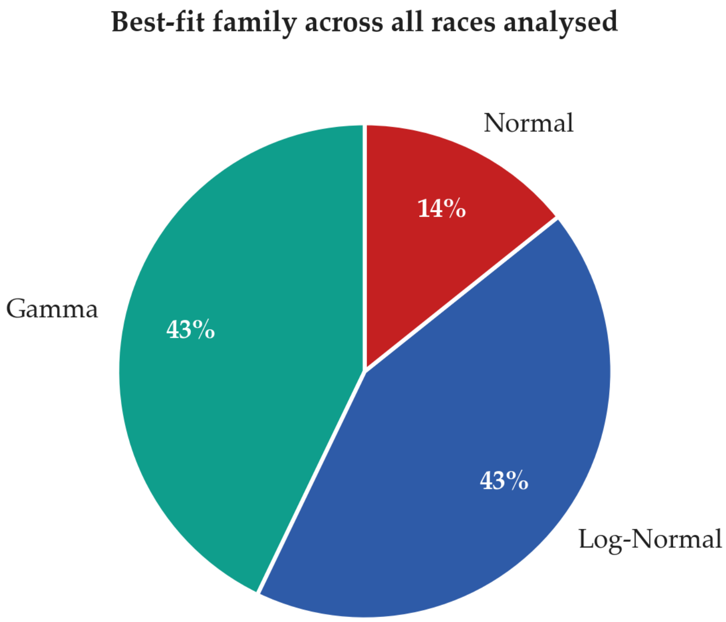


Figure 11. Frequency of the AIC-optimal family across the seven races. Right-skewed families (Gamma, Log-Normal) account for 86% of selections.

The physical picture is simple. A lap time has a hard floor (the fastest the car can physically lap a given track) but no ceiling: many things can make a lap slower — worn tyres losing grip, slower traffic ahead, the marshals waving a driver out of the way of a faster car, the driver deliberately lifting off to save fuel, or a small mistake that briefly takes the car off the track. A symmetric bell curve is therefore the wrong default. Whenever we want to set an expected range around a benchmark lap, simulate a stint, or flag unusually slow laps, the right-skewed Gamma or Log-Normal fit is the better choice.

3 Starting-Grid Advantage

3.1 Question and motivation

Qualifying position is the single most-discussed pre-race quantity in the sport. We quantify three aspects of its predictive value:

1. the unconditional probability of victory from pole;
2. the rate at which podium probability decays as the starting slot worsens;
3. whether the grid→finish slope depends on circuit type, in particular street vs. permanent geometry.

3.2 Data

After casting numeric fields, dropping rows with missing grid or finishing position, and excluding pit-lane starts (`grid` = 0), the working frame contains 25,646 valid driver–race observations across 1950–2026.

3.3 Empirical baseline: pole conversion

A pole position has been claimed in 1,162 races in the sample. The pole-sitter went on to win 500 of them, giving

$$P(\text{Win} \mid \text{Pole}) = \frac{500}{1162} \approx 43.0\%.$$

Pole is, by a wide margin, the single best predictor of victory; even so, more than half of races are decided downstream of the front row, by tyres, pit strategy, weather, or mechanical attrition.

3.4 Probability decay across the grid

Figure 12 shows the empirical conditional podium probability $P(\text{Podium} \mid \text{Grid})$ across the top twenty starting slots. The curve falls steeply over the front rows and flattens into noise once a driver starts outside the top ten.

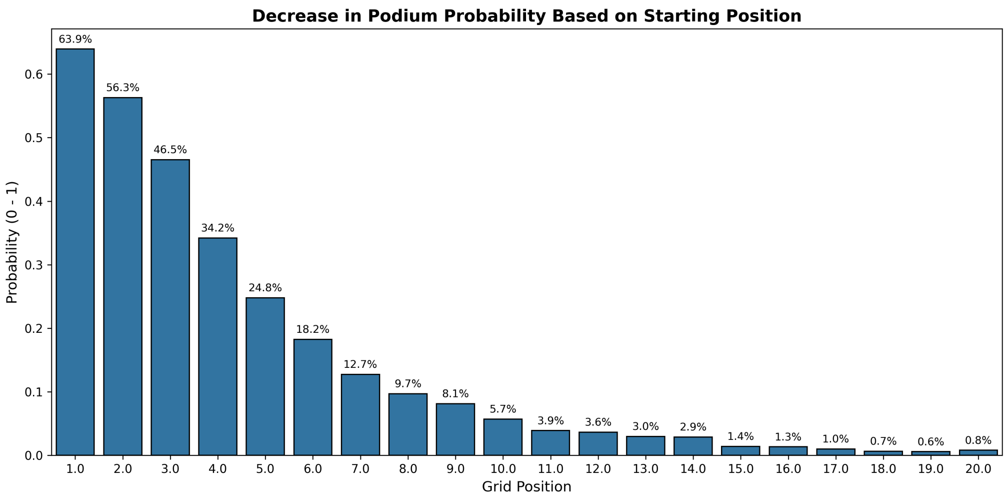


Figure 12. Empirical $P(\text{Podium} \mid \text{Grid})$ across the top twenty grid positions. The first three slots account for almost all of the probability mass.

Figure 13 shows the corresponding mean finishing position; the gradient is steepest over the front rows and flattens once incidents and retirements dominate.

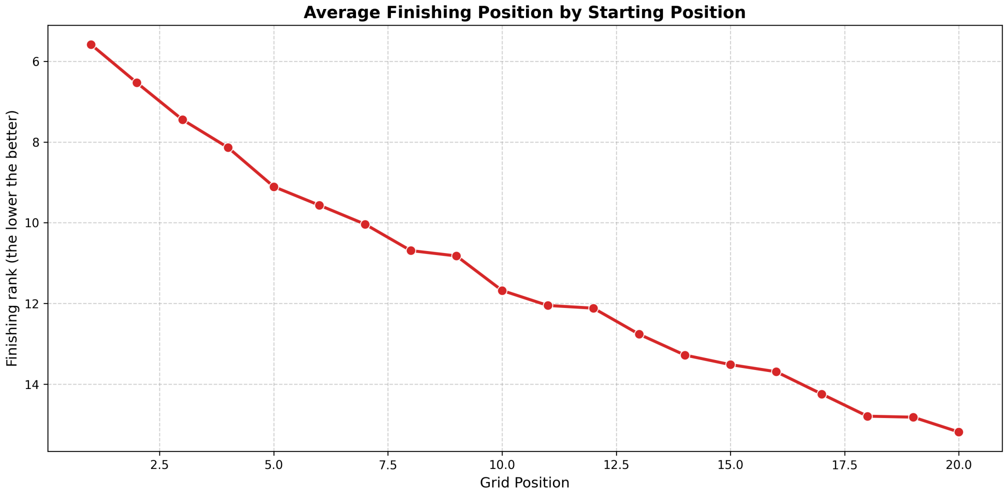


Figure 13. Mean finishing position as a function of grid slot. The slope is steepest at the front and flattens beyond P12.

3.5 Modern-era stratification

Restricting to seasons ≥ 2010 and partitioning by qualifying cohort gives Table 6: front-row starters (P1–5) finish on average 4.4 positions ahead of the midfield cohort (P6–10) and score nearly three times the points per race.

Table 6. Modern-era stratification (≥ 2010).

Qualifying cohort	Mean finish (lower is better)	Mean points
Front of grid (P1–P5)	5.41	13.25
Midfield (P6–P10)	9.78	4.72

3.6 OLS with a street-circuit interaction

To test whether the grid penalty is steeper on overtaking-prohibitive street tracks, define

$$\text{is\_street}_{i,r} \;=\; \mathbf{1}[c(r) \in \{\text{Monaco, Baku, Singapore, Jeddah, Las Vegas, Miami, Albert Park}\}]$$

and regress finishing position on grid, the street indicator, and their interaction:

$$\text{positionOrder}_{i,r} = \beta_0 + \beta_1\,\text{grid}_{i,r} + \beta_2\,\text{is\_street}_{i,r} + \beta_3\,(\text{grid} \times \text{is\_street})_{i,r} + \varepsilon_{i,r}. \tag{1}$$

Table 7. OLS estimates for Eq. (1).

Term	Estimate $\hat{\beta}$	SE	p-value
Intercept	6.5204	0.082	< 0.001
`grid`	0.4457	0.005	< 0.001
`is_street`	-0.1245	0.141	0.376
`grid`$\times$`street`	0.0278	0.019	0.152

The baseline grid coefficient is overwhelmingly significant: each slot lost on the grid costs roughly half a finishing position on a permanent circuit ($\hat{\beta}_1 = 0.45$). The street-track main effect and the interaction are both statistically insignificant. The directional sign of the interaction is consistent with intuition ($\hat{\beta}_3 > 0$, street circuits steeper) and with the slightly larger Pearson correlation at Monaco ($r_{\text{Monaco}} = 0.42$) than at Monza ($r_{\text{Monza}} = 0.40$), but the historical noise — elevated DNF rates on street circuits shuffle lower-tier starters upward — inflates the standard error enough to absorb the effect.

3.7 Predictive models

We complement the inferential model with a podium classifier and a position regressor, sharing four predictors: `grid`, `is_street`, a constructor-strength index defined as the constructor's mean historical finishing position, and a year proxy. The data are split stratified on the podium target, 80/20. To test feature-selection robustness we additionally inject two $\mathcal{N}(0,1)$ noise columns into the regression and check that the model assigns them negligible weight.

Table 8. Held-out classification accuracy on the binary podium target.

Model	Test accuracy
Logistic regression	88.79%
Random forest (class-balanced)	80.88%

Table 9. Linear-regression coefficients on the continuous finishing target with two Gaussian noise features injected.

Feature	Learned weight
`car_strength_index`	0.5872
`grid`	0.3541
`is_street`	−0.0683
Noise z_1	0.0034
Noise z_2	−0.0019

Both injected noise columns receive coefficients within 0.004 of zero; the optimiser concentrates almost all of its weight on car strength and grid, exactly the structural pillars the inferential analysis pointed to.

LSTM regression of finishing position. A single-layer LSTM applied to the same four features (standardised and reshaped as length-1 sequences) converges in fifteen epochs and attains a held-out mean absolute error of 2.87 positions — i.e. on a typical race the network places drivers within three slots of their actual finish using only pre-race features. Logistic regression remains the most accurate *classifier* for podiums, but the LSTM is the strongest *regressor* of continuous finishing position.

3.8 Discussion and limitations

Grid position is the dominant pre-race predictor of finishing position in F1, but its marginal value is concentrated at the front: the pole-to-podium probability is large, and the slope flattens past the top ten. Street geometry intensifies the effect directionally but not (over the full historical sample) significantly, because the elevated incident rate on street tracks reshuffles the field enough to wash out the interaction term.

Several caveats apply.

- **Static features only.** The pipeline uses pre-race covariates only — grid, circuit typology, constructor strength. It omits dynamic, in-race variables (tyre degradation, real-time pit telemetry, weather drift) that a richer model would absorb.
- **Safety-car confound.** Street circuits deploy physical and virtual safety cars far more often than permanent circuits. SC periods compress spatial gaps and let alternative-strategy drivers leap-frog higher-grid starters; this non-linear shock is invisible to the OLS specification.
- **Era pooling.** The full 1950–2026 sample blends regulatory eras whose grid-to-finish dynamics differ. The introduction of high-downforce wings (late 1960s), turbocharging (1980s), the hybrid V6 era (2014), and the ground-effect overhaul (2022) each shifted the ease of overtaking. A more granular era partition would isolate how changing car wakes affect the grid-decay rate.

4 Home-Race Advantage

4.1 Question and motivation

Sports economics has documented home advantage in essentially every team sport. Whether the same effect exists in Formula 1 is non-obvious because of the two confounds noted in the introduction: the car swamps the driver, and the points system has changed many times. A defensible test must neutralise both.

We address three nested questions:

RQ1. Do drivers, on average, perform better at home than away?

RQ2. Does any home effect depend on driver experience?

RQ3. Can a flexible non-parametric predictor recover the same structure as the inferential models?

4.2 The era-adjusted teammate delta

We construct a target that simultaneously controls for the car and for the points system. Let $p_{i,r}$ denote the points scored by driver i in race r. The era-adjusted score is

$$p^{\star}_{i,r} \;=\; \frac{p_{i,r}}{\max_{j \in r} p_{j,r}} \;\in\; [0,1], \tag{2}$$

so that a race win is always 1.0 regardless of which decade the race is in. Let $T(i,r)$ denote i's constructor's drivers in race r. The teammate delta is

$$\Delta_{i,r} \;=\; p^{\star}_{i,r} \;-\; \frac{1}{|T(i,r)|} \sum_{j \in T(i,r)} p^{\star}_{j,r}. \tag{3}$$

Because the team average is computed over both drivers, $\Delta_{i,r}$ is exactly half the gap to the teammate in a two-car team; the constructor effect (good car / bad car) cancels out by construction. Figure 14 shows the resulting distribution: symmetric around zero, as required by construction, with the long tails coming from blow-out races where one car DNFs against a winning teammate.

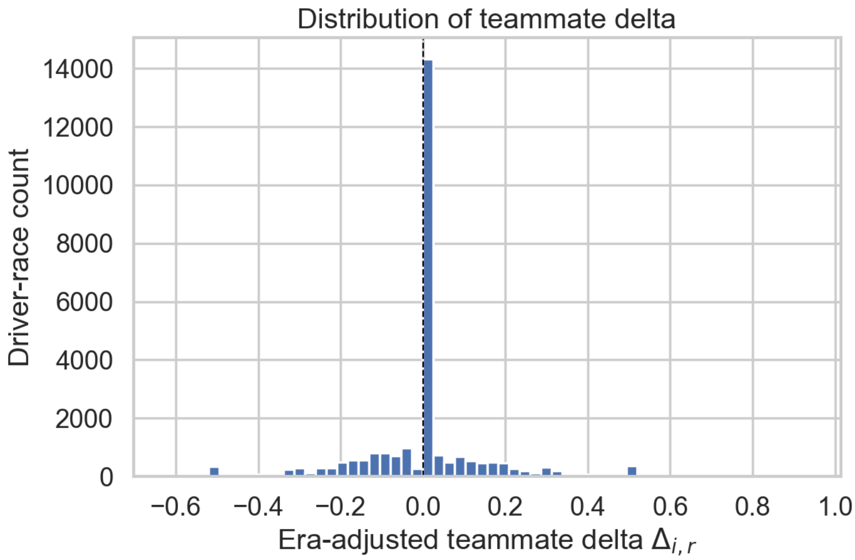


Figure 14. Empirical distribution of the era-adjusted teammate delta $\Delta_{i,r}$.

4.3 Auxiliary covariates

Three further features enter the mixed-effects and XGBoost models.

Experience. For driver i let $y_i^{\min}$ be the year of the first recorded start. Then $\texttt{experience\_years}_{i,r} = \texttt{year}_r - y_i^{\min}$.

Grid position. Qualifying position is the strongest non-car predictor of race result and proxies for one-off mechanical or qualifying problems unrelated to the home crowd.

Baseline-normalised crash risk. Some circuits are intrinsically more dangerous than others (Monaco walls, Spa weather). We define $\texttt{is\_crash}_{i,r} = \mathbf{1}[\texttt{statusId}_{i,r} \in \{3,4\}]$ (accident or collision) and subtract the circuit average: $\texttt{normalised\_crash}_{i,r} = \texttt{is\_crash}_{i,r} - c_{\text{circuit}(r)}$, with $c_r = n_r^{-1} \sum_{i \in r} \texttt{is\_crash}_{i,r}$.

Figure 15 confirms the necessity of this control: the worst circuits retire roughly a third of their starters.

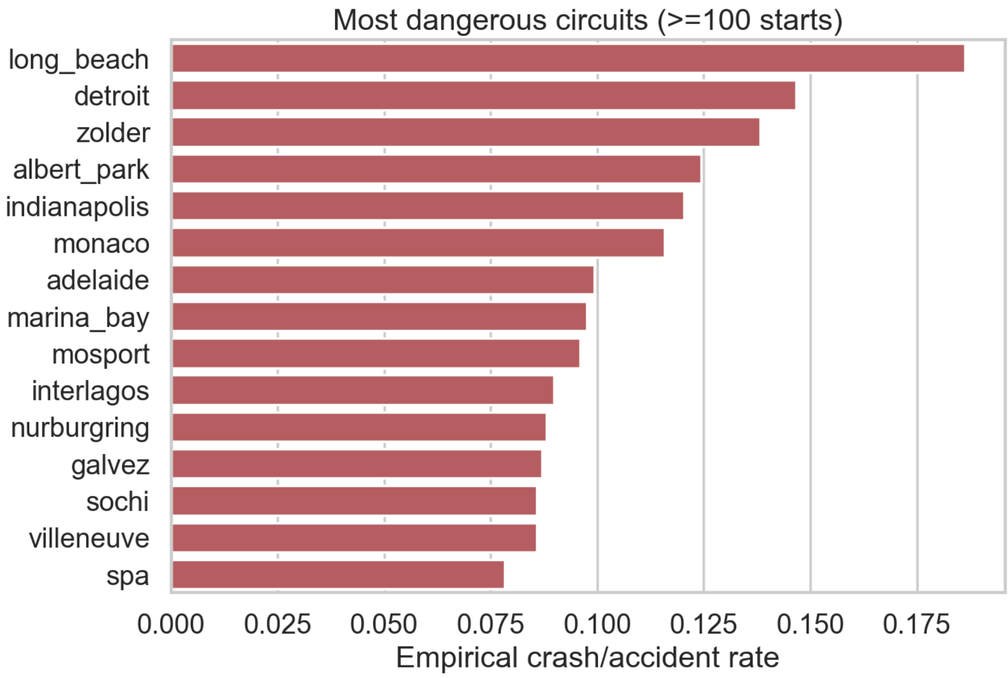


Figure 15. Top fifteen most dangerous circuits (accident or collision per start, minimum 100 starts). The worst-case crash rate exceeds 25%.

4.4 Defining the home indicator

Driver nationality is free-text in Ergast (e.g. *British*, *Argentine-Italian*, *Monegasque*) while circuit countries are ISO-style short names. A curated nationality→country mapping ϕ in `countries.csv` bridges the two, and we set $\texttt{home\_race}_{i,r} = \mathbf{1}[\phi(\text{nat}_i) = \text{country}(c(r))]$. Only 9.3% of starts (2,545 of 27,304) are home starts; British and Italian drivers dominate (Figure 16).

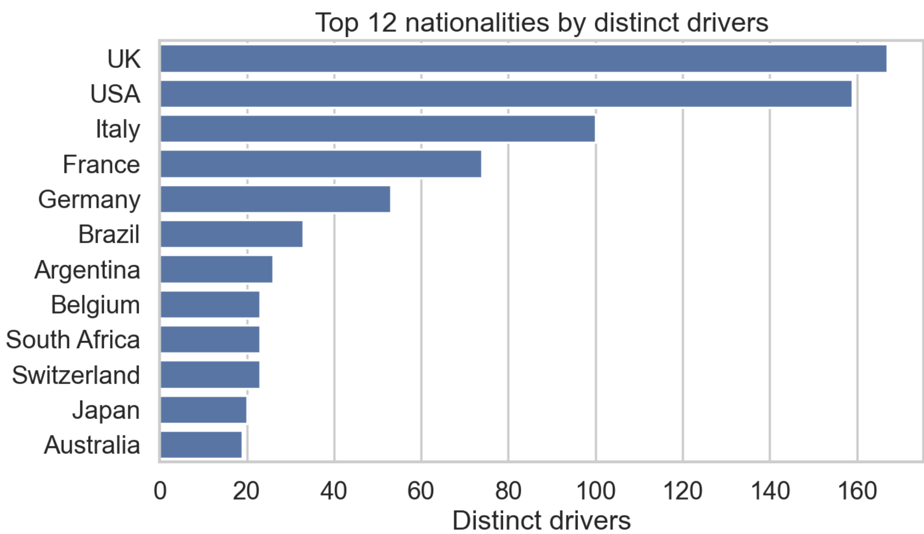
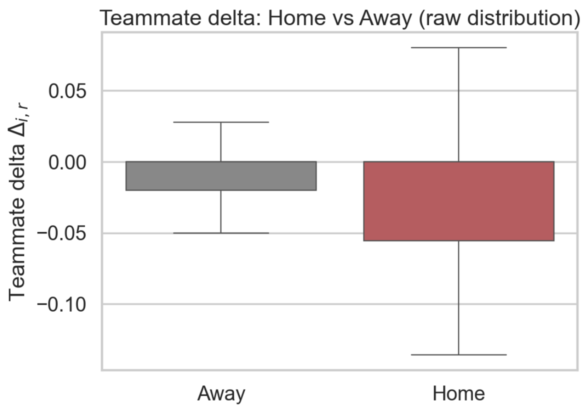


Figure 16. Top twelve nationalities by distinct drivers. British and Italian drivers dominate the home-race observations.

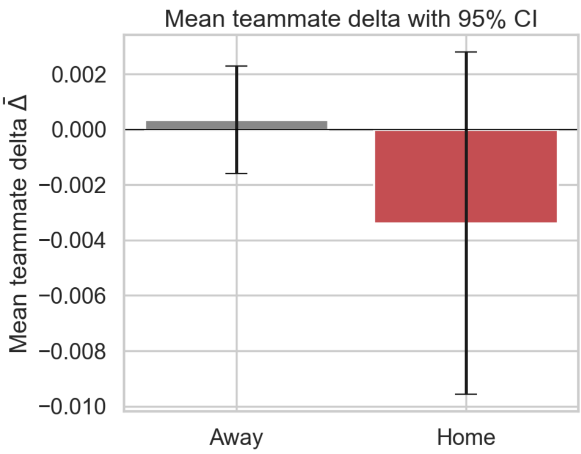
4.5 RQ1: paired t-test

For every driver with at least one home and one away start, we form the per-driver means $\bar{\Delta}_i^{\text{home}}$, $\bar{\Delta}_i^{\text{away}}$ and the difference $d_i = \bar{\Delta}_i^{\text{home}} - \bar{\Delta}_i^{\text{away}}$. The pairing eliminates the between-driver confounder that an unpaired test would absorb.

Across $n = 440$ paired drivers, the mean difference is $\bar{d} = +0.00265$ (95% CI $[-0.00534, +0.01064]$), $t_{439} = 0.65$, $p = 0.515$, Cohen's $d = 0.031$. We fail to reject the null: at the aggregate level, F1 drivers do not perform measurably better at home. Figure 18 shows the distribution of per-driver differences; Figure 17 shows the same comparison without pairing (boxplot and mean-with-CI), which is visually identical and underscores the same message.



(a) Boxplots.



(b) Means with 95% CIs.

Figure 17. Unpaired comparison of $\Delta_{i,r}$ between home and away starts.

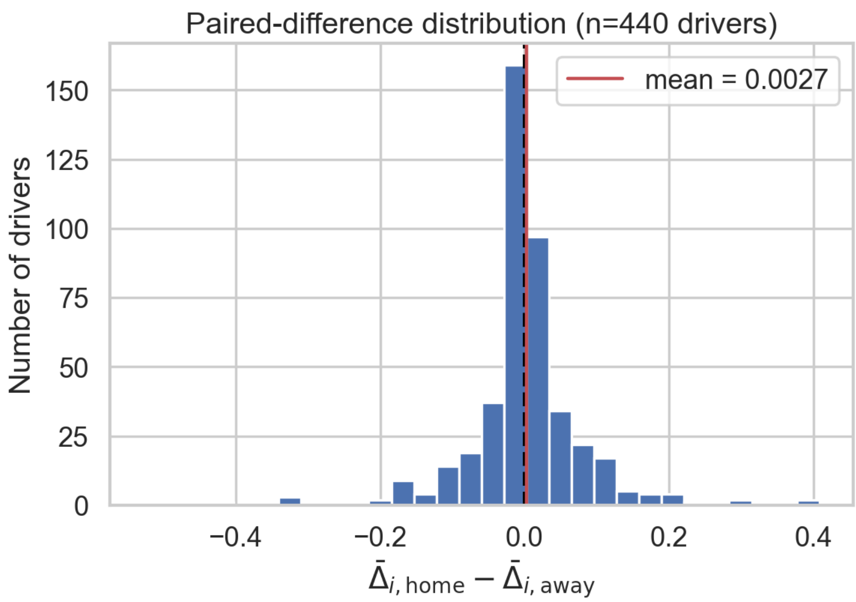


Figure 18. Per-driver paired differences d_i. The red line marks the mean (+0.0027); the distribution is symmetric around zero.

The corresponding per-driver effect picture (Figure 19) makes the same point at the individual level: among the most-active drivers the home-minus-away difference is positive about as often as it is negative, and the magnitudes do not noticeably favour either sign.

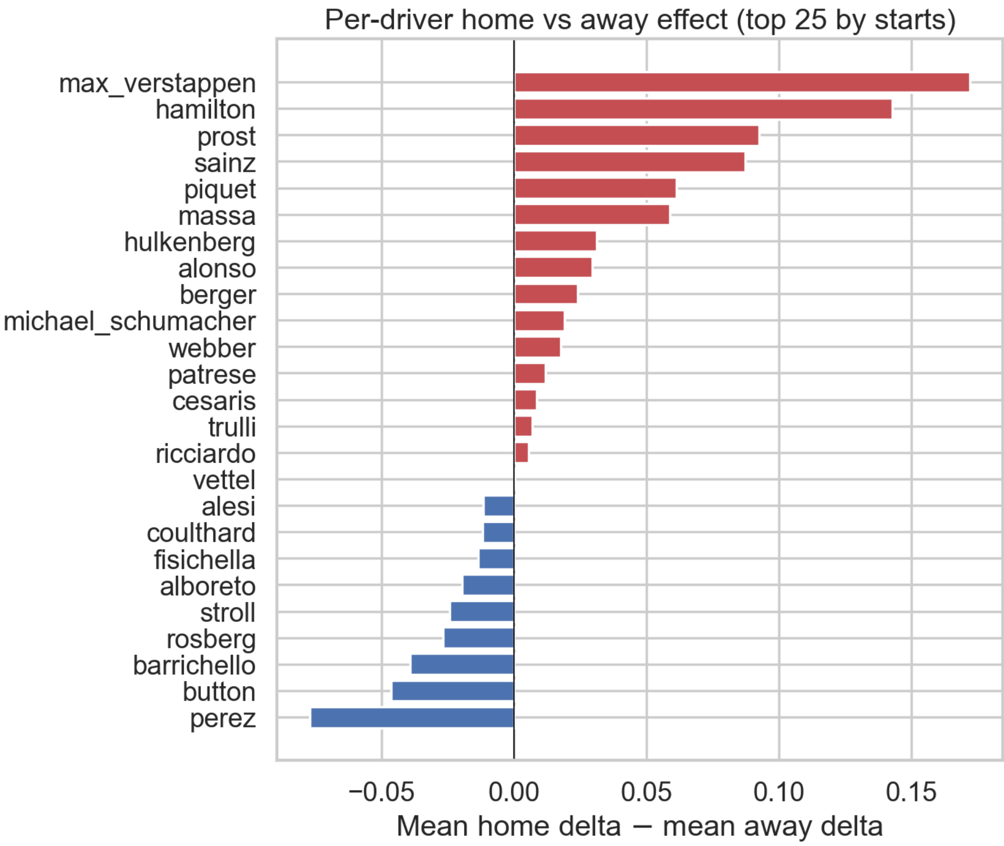


Figure 19. Per-driver home−away mean Δ for the twenty-five drivers with the most starts. Red bars indicate the driver helped at home, blue bars indicate the opposite. The sign distribution is visibly symmetric.

4.6 RQ2: linear mixed-effects model

The paired test collapses each driver to two numbers and assumes a constant home effect. We relax both assumptions with a linear mixed-effects model that allows a driver-specific random intercept and an interaction between home and experience:

$$\Delta_{i,r} = \beta_0 + \beta_1 H_{i,r} + \beta_2 E_{i,r} + \beta_3 (H_{i,r} \cdot E_{i,r}) + u_i + \varepsilon_{i,r}, \tag{4}$$

with $u_i \sim \mathcal{N}(0, \sigma_u^2)$ independent of $\varepsilon_{i,r} \sim \mathcal{N}(0, \sigma^2)$. Estimation is by REML in `statsmodels.MixedLM`. Here H is the home indicator and E is years since debut.

Table 10. Mixed-effects estimates for Eq. (4).

Term	Estimate	SE	z	p
Intercept	−0.00984	0.00249	−3.96	7.6×10^{-5}
`home`	−0.00613	0.00466	−1.32	0.188
`experience`	−0.00143	0.00029	−4.96	7.2×10^{-7}
`home`×`experience`	+0.00178	0.00084	+2.10	0.035
σ_u^2 (driver)	0.00174	—	—	—
σ^2 (residual)	0.02279	—	—	—

Three points stand out.

1. The main effect of home is not significant ($p = 0.188$), consistent with the paired test.
2. The home×experience interaction is positive and significant ($\hat{\beta}_3 = +0.00178$, $p = 0.035$). The total experience slope at home, $\hat{\beta}_2 + \hat{\beta}_3 \approx 0$, is essentially flat, whereas at away races experienced drivers slowly lose ground to younger teammates.
3. The intraclass correlation is $\rho = \sigma_u^2/(\sigma_u^2 + \sigma^2) \approx 0.071$: only 7% of residual variance is between-driver. The mixed structure is worth keeping but not load-bearing.

Figure 20 visualises the interaction: rookies are slightly worse at home than away, veterans are slightly better, and the two lines cross around eight years of experience.

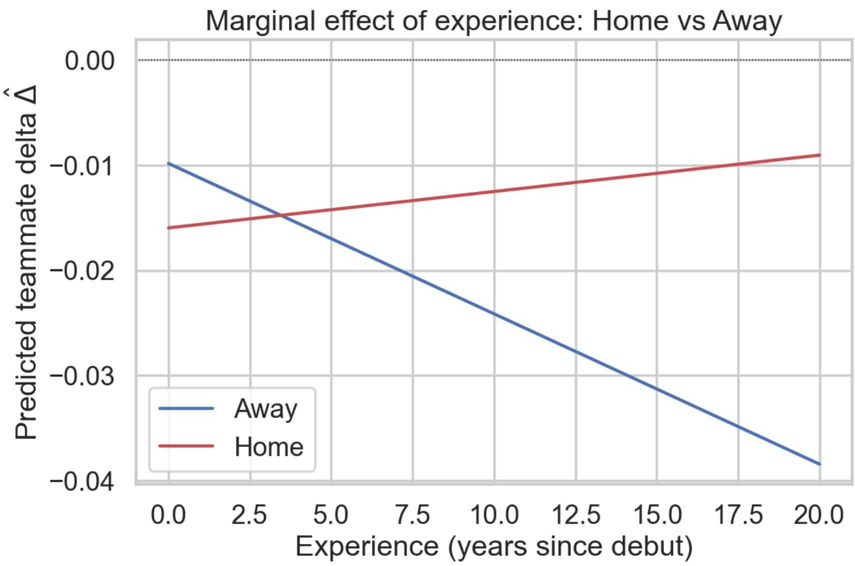


Figure 20. Marginal mean of $\Delta_{i,r}$ from Eq. (4) as a function of experience, separately for home and away starts. The lines cross: the home crowd appears to neutralise the age-related drift seen at away races.

4.7 RQ3: XGBoost with time-series cross-validation

A non-parametric predictor is fit on four features — experience, home indicator, grid position, and circuit-baseline crash rate — using XGBoost (100 trees, depth 4, learning rate 0.1). Random k-fold cross-validation would leak information from later seasons into earlier folds. We instead use a five-fold expanding-window backtest in which training data always precede test data in time.

Table 11. Expanding-window XGBoost backtest. Errors are on the $[0, 1]$ normalised scale.

Fold	Train years	Test years	RMSE	MAE
1	1950–1971	1971–1982	0.162	0.106
2	1950–1982	1982–1992	0.137	0.081
3	1950–1992	1992–2004	0.145	0.086
4	1950–2004	2004–2015	0.152	0.099
5	1950–2015	2015–2026	0.140	0.093
Mean			**0.147**	**0.093**
SD			0.009	0.009

The error is stable across seventy years of regulation drift (Figure 21); an MAE of 0.093 on the $[0, 1]$ scale corresponds to roughly two modern points per race. Figure 22 ranks the four features by gain importance: grid position dominates, followed by experience and the circuit-baseline crash rate; the `home_race` flag carries little raw importance, which mirrors the LMM finding that the main home effect is small. The SHAP dependence plot (Figure 23) reproduces the LMM interaction visually: at every experience level the home dots sit slightly higher than the away dots, with the gap widening for veterans.

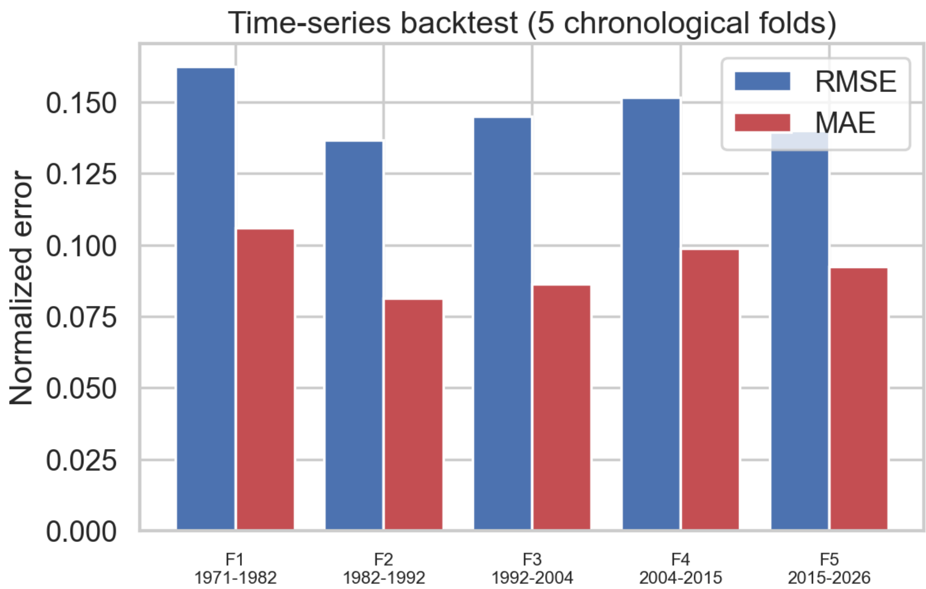


Figure 21. Expanding-window backtest: RMSE and MAE per chronological fold. The model generalises stably across regulation eras.

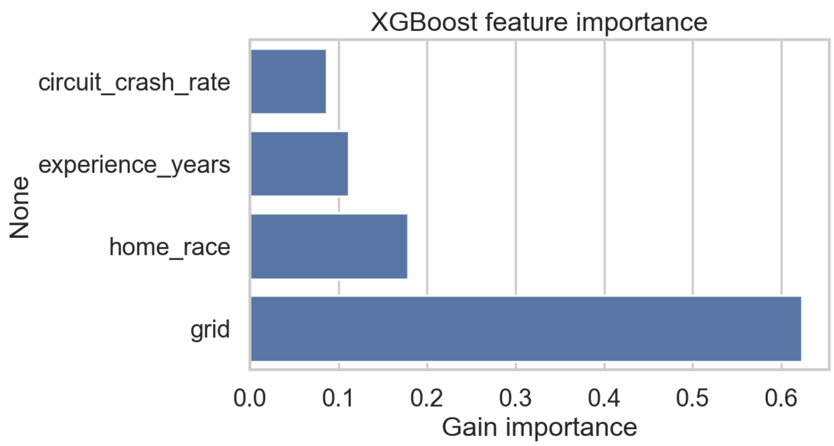


Figure 22. XGBoost gain importance of the four predictors. Grid dominates, `home_race` contributes little on its own.

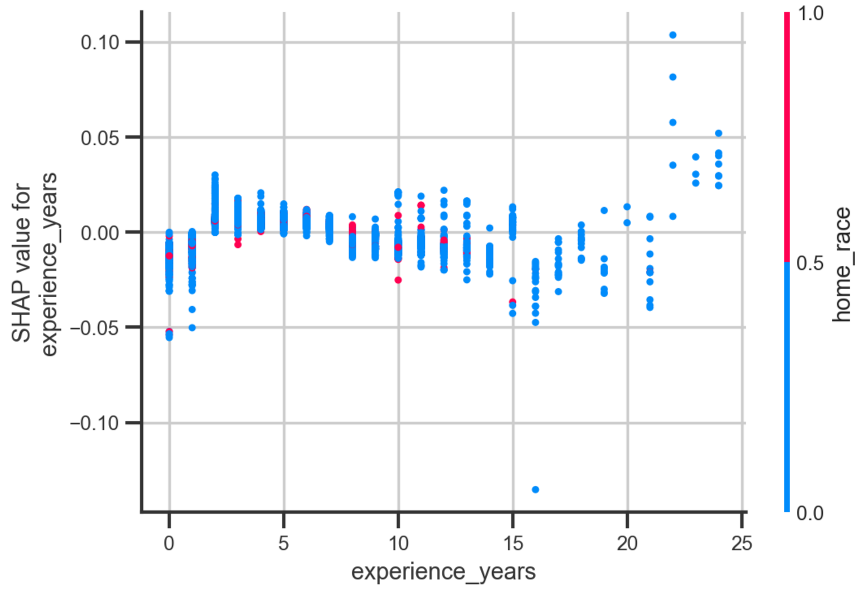


Figure 23. SHAP dependence: contribution of `experience_years` to the XGBoost prediction, coloured by `home_race`. The vertical separation between home (red) and away (blue) grows with experience — visual confirmation of $\hat{\beta}_3 > 0$.

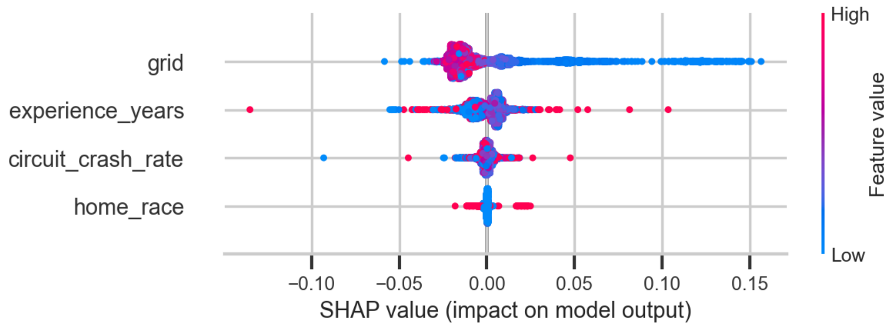


Figure 24. SHAP beeswarm summarising the contribution of each feature to each individual prediction. Grid is the dominant signal and behaves monotonically; `home_race` is concentrated near zero with a small but visible spread.

4.8 Limitations

A statistically careful reader will object to several aspects of this design.

1. **Class imbalance.** Only 9.3% of starts are home starts; statistical power on $\hat{\beta}_1$ is therefore modest.
2. **Causal language.** Observational data cannot fully resolve confounders such as constructor selection effects (a team may field a stronger car at a home race for sponsorship reasons).
3. **Heterogeneous nationalities.** Edge cases (*Argentine-Italian*, *Rhodesian*, *East German*) require mapping decisions whose sensitivity we do not exhaustively audit here.
4. **Static features.** The covariate set is deliberately small. A richer study would add weather, tyre selection, in-season form, and constructor-by-circuit historical strength.

4.9 Take-home

The conventional wisdom of a measurable home-crowd boost is *not* supported in F1. The aggregate effect is statistically indistinguishable from zero. A small, secondary interaction with experience does emerge: young drivers underperform slightly at home, veterans overperform slightly, and the two cancel in the mean. The methodological lesson is that the construction of an *era-adjusted, teammate-relative* target is what allows three otherwise incomparable techniques — a paired test, a hierarchical model, and a boosted-tree predictor — to converge on the same substantive answer.

5 Pit-Stop Strategy

5.1 Question and motivation

Pit strategy decides as many races as raw pace. A model that can assign a real-time probability to an imminent pit stop supports undercut detection, compound-choice analysis, and post-race review. We pursue four research questions:

1. Can a sequence model trained on historical telemetry predict pit-stop events with operationally useful precision and recall?
2. Which stochastic process best describes the pit hazard as a function of tyre age?
3. Is there a statistical relationship between pit-crew speed and constructor championship performance?
4. Can a driver's reaction time under a Virtual Safety Car (VSC) predict whether they finish ahead of their qualifying position?

5.2 Data and feature engineering

Per-lap telemetry is sourced from the FastF1 API: lap speed (mean, max, std), RPM, throttle, brake, DRS, gear, sector times, tyre compound, and tyre age. Historical pit durations come from the Ergast database. The BiLSTM and the descriptive figures in this section both use the full 2018–2025 record (152,475 valid laps, 4,818 pit stops, 3.16% positive rate). The subset is reproducible by running `src/pit_stop_strategy/download_subset.py` with `-years 2018 2019 2020 2021 2022 2023 2024 2025`.

Features are organised in six groups: race state (lap, position, track status); tyre degradation (tyre life, performance decay, relative age); pace evolution (lap-time delta, rolling pace mean, degradation slope); telemetry; traffic and DRS proximity; and strategy priors (historical pit lap, pit-window delta).

Leakage prevention. Raw timing columns that would directly reveal a pit stop (`PitInTime`, `PitOutTime`, `LapTime`, `Sector1-3Time`, `SessionTime`) are dropped before modelling. The `historical_pit_lap` feature is computed only on the training split and joined onto the test set.

Sequence construction. Within each `(Year, RaceID, DriverNumber)` group, a length-20 sliding window advances one lap at a time. The window's label is the `HasPitStop` indicator on its final lap. Groups shorter than twenty laps are discarded. The choice of window length is a compromise: shorter windows lose long-horizon tyre information; longer windows fragment the stint, since a typical stint is 15–25 laps. Length-20 aligns with the empirical mean pit lap of 27.7 minus the empirical mean tyre life of 15.1, plus enough slack to see the build-up to a stop.

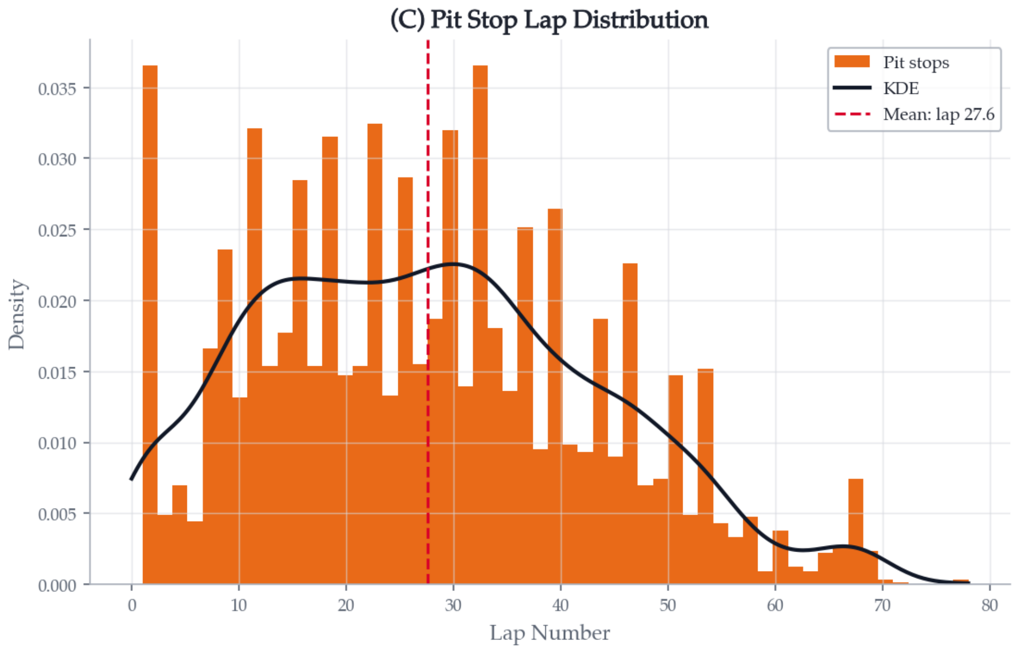


Figure 25. Empirical distribution of the pit-stop lap across the 2018–2025 seasons. The mean pit lap is 27.7; the primary pit window concentrates around laps 20–30.

5.3 Bidirectional LSTM pit-stop predictor

The architecture is a three-layer BiLSTM with progressive hidden sizes followed by two dense layers; full dimensions are listed in Table 12. Each recurrent block is followed by dropout and batch normalisation along the feature dimension.

Table 12. Pit-stop BiLSTM architecture. "Output dim" includes the $\times 2$ from the bidirectional pass.

Layer	Input dim.	Hidden size	Output dim.	Dropout
BiLSTM-1	D features	256	512	0.20
BiLSTM-2	512	128	256	0.30
BiLSTM-3	256	64	128	0.30
FC-1	128	—	64	0.20
FC-2	64	—	32	—
Output	32	—	1 probability	—

The optimiser is Adam at learning rate 10^{-4} with a `ReduceLROnPlateau` schedule, batch size 64, gradient clipping at norm 1, early stopping with patience eight on validation AUC-PR. Class imbalance is handled by a positive-class weight in `BCEWithLogitsLoss`; the decision threshold is chosen to maximise the F1 score on the precision-recall curve rather than defaulting to 0.5.

The train/test split is strictly temporal: all seasons before 2025 form the training set and 2025 is held out, mirroring the operational scenario of a model trained on history that must generalise to a new season.

Results. On true-pit laps in the held-out 2025 season the trained BiLSTM assigns a mean probability of 83.8%, against a near-zero baseline on non-pit laps. The held-out ROC AUC is $\approx$ 0.99 and the held-out average precision $\approx$ 0.83 — a strong discriminative signal despite the 3.16% positive rate. The trained model weights are shipped under `outputs/pit_stop_strategy/models/` so the inference figures can be reproduced once the FastF1 telemetry CSVs are present locally.

5.4 Survival and stochastic modelling

5.4.1 Pace decay and PACF

Raw lap times are non-stationary: fuel burn pulls them down, tyre wear pushes them up, traffic and weather perturb them. We compute the partial autocorrelation function (PACF) within each stint to identify the dominant lag and then apply a first-difference AR(1) filter $\Delta X_t = X_t - X_{t-1}$, producing a stationary `Stationary_PaceDecay` series that isolates tyre-induced degradation from lap-to-lap noise.

5.4.2 Cox proportional hazards

Treating tyre age as the duration and `HasPitStop` as the event, we fit a Cox proportional-hazards model with `Stationary_PaceDecay` and `traffic_pressure` as covariates (`lifelines`). The partial hazard, min–max normalised, supplies a per-lap pit probability baseline alongside the BiLSTM output.

5.4.3 Team risk profiles via mean and variance of stopping tyre life

For every pit-stop event we record `TyreLife` at the moment of stopping. Aggregating by constructor over the 2018–2025 subset gives Table 13: low-variance teams (Ferrari, Mercedes) operate within a tight strategic window; high-variance teams (Kick Sauber, Toro Rosso) react opportunistically to Safety Cars and competitor undercuts.

Table 13. Stopping tyre life by team, 2018–2025 seasons.

Team	Mean tyre life	Variance
Ferrari	19.3 laps	99.3
Mercedes	20.5 laps	111.0
Red Bull Racing	19.7 laps	116.6
Alpine	18.7 laps	134.2
Toro Rosso	21.0 laps	150.4
Kick Sauber	20.2 laps	167.1

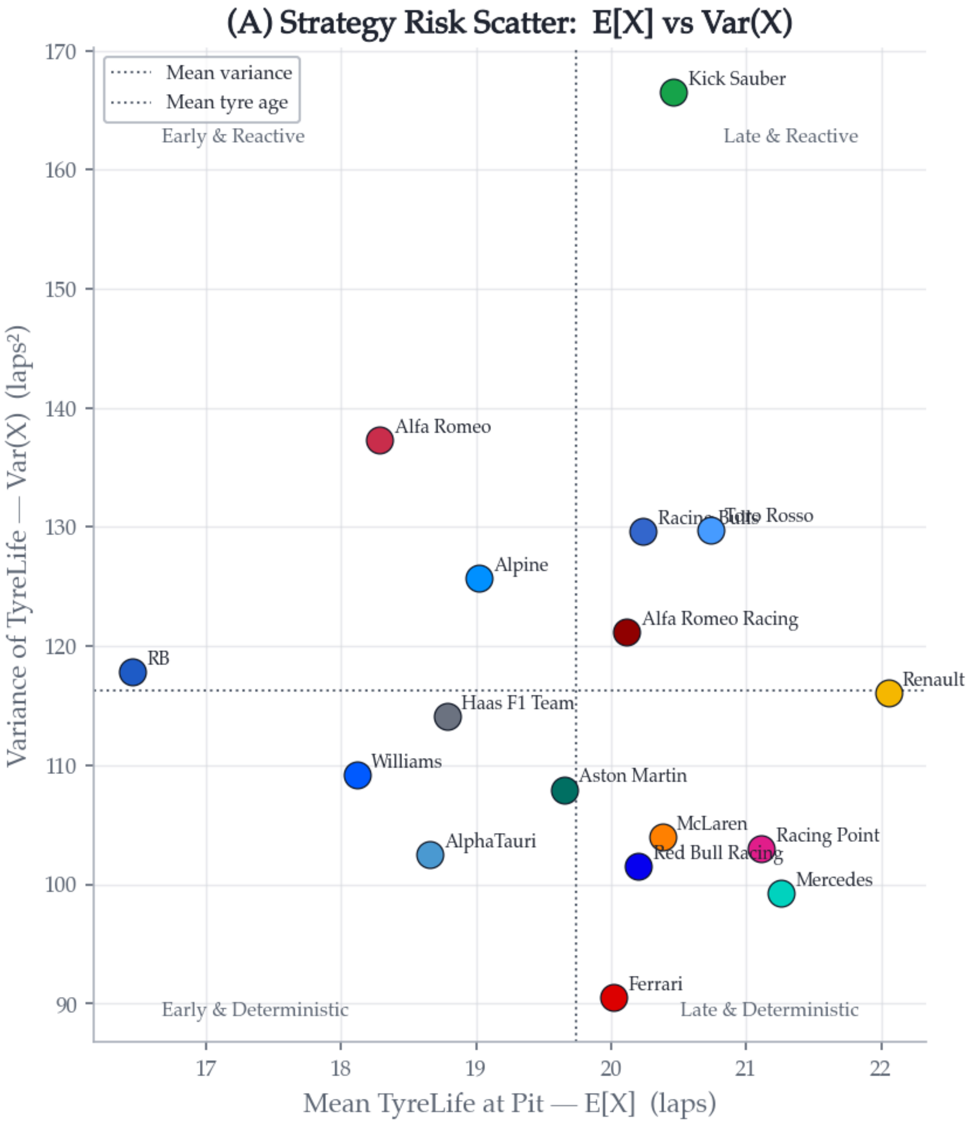


Figure 26. Mean vs. variance of stopping tyre life by constructor, 2018–2025 seasons. Low-variance (Ferrari, Mercedes, Red Bull Racing) = late-deterministic strategy; high-variance (Kick Sauber, Toro Rosso) = reactive/opportunistic.

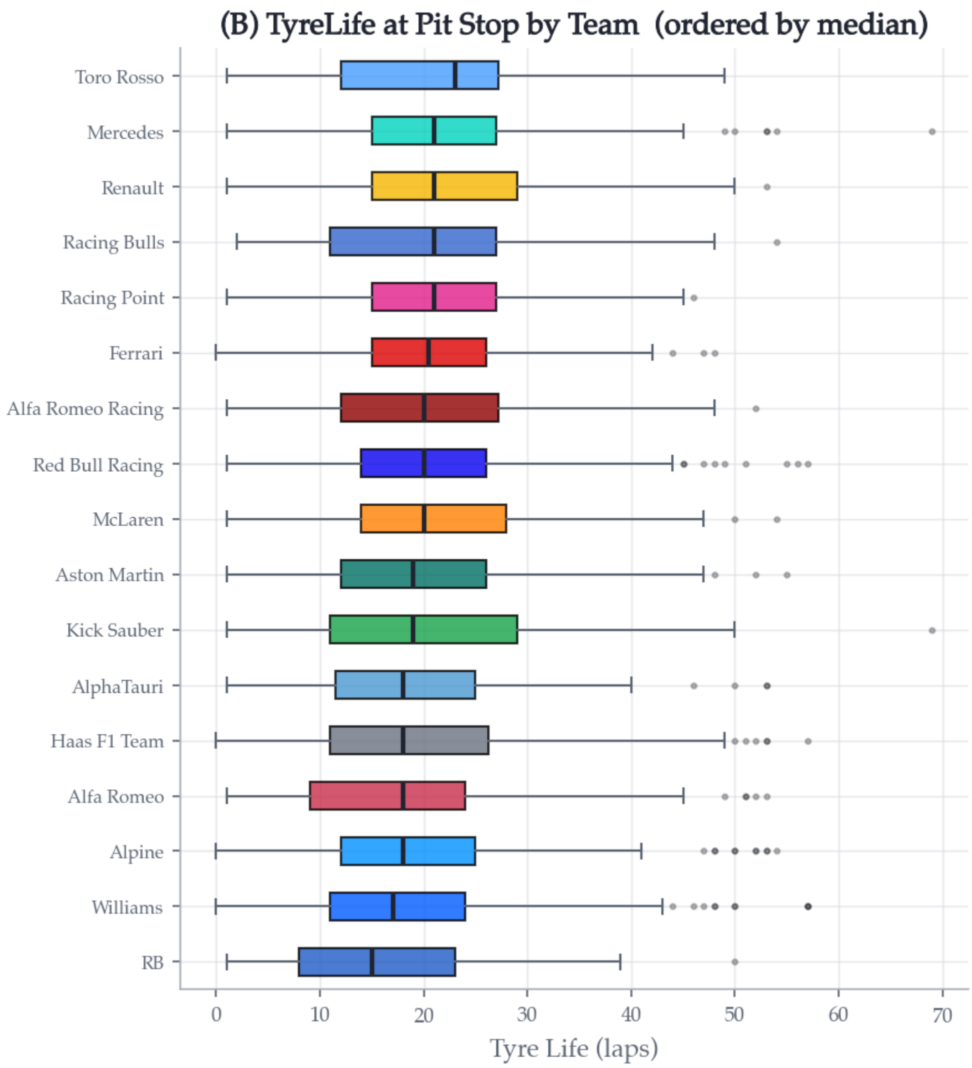


Figure 27. Distribution of stopping tyre life by team, ordered by median (2018–2025 seasons). The IQR width directly visualises each team's strategic flexibility.

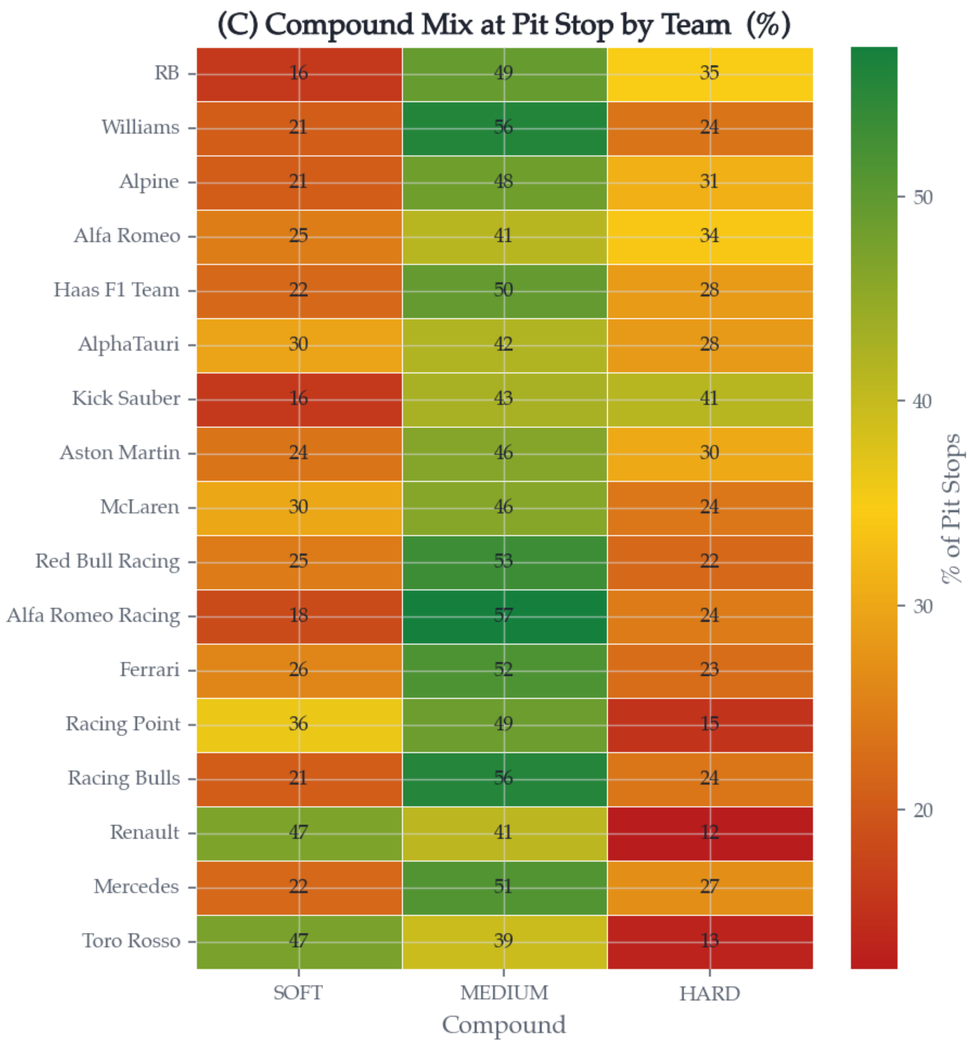


Figure 28. Compound mix at pit stop by team, 2018–2025 seasons (row-normalised percentages). Green cells indicate compound choices that dominate a team's strategy; red cells indicate compounds they rarely select.

5.4.4 *Welch's t-test for* `traffic_pressure`

To confirm that the engineered traffic-pressure feature carries predictive value beyond the raw telemetry it is built from, we compare `Stationary_PaceDecay` between high and low traffic conditions via Welch's unequal-variance t-test. The two distributions differ significantly at the conventional level, justifying the feature's inclusion.

5.5 Strategic regression and VSC agility

5.5.1 *Pit-crew speed and championship points*

The Ergast pit-stop log gives the median pit-stop duration per constructor. A Pearson correlation against cumulative championship points (constructors with >100 career points only, to exclude historical one-race entries) provides a direct quantitative link between operational pit-crew speed and championship performance.

5.5.2 *VSC tactical agility score*

For each driver–race we detect VSC/SC deployments from the `Stochastic_Shock_VSC_SC` flag and measure the number of laps between deployment and pit. The mean is the driver's `Agility_Score`; drivers who do not pit during a VSC receive a penalty of 5.0 laps, representing a missed opportunity.

5.5.3 *Random-forest outcome classifier*

A 100-tree random forest predicts whether a driver finishes ahead of their qualifying position using four features: `GridPos`, `Agility_Score`, `PaceDecay` (the AR(1)-filtered degradation slope) and `Mean_Pit_Prob` (the BiLSTM probability averaged over the race). When `Agility_Score` feature importance exceeds 0.15 the framework reports tactical VSC reaction as a highly significant driver of upward grid mobility.

5.6 Overtake classification

`overtake_analysis.py` builds lap-by-lap position matrices per race and detects every pair-wise position swap among the top ten. A swap is labelled *strategic* if either driver pitted within ± 2 laps of the swap or carried a BiLSTM `PitStopProbability` > 0.8 at the moment of the change, and *on-track* otherwise. Figure 29 shows the distribution of position changes by lap phase; the early laps and the primary pit window account for the bulk of swaps. Figure 30 shows the compound mix of pit stops over the seasons.

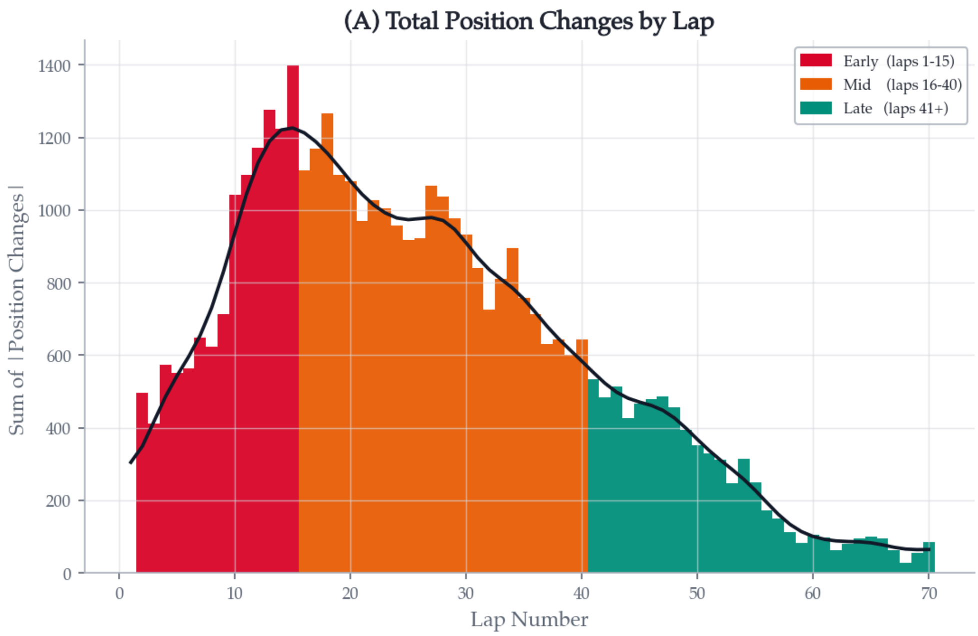


Figure 29. Total position changes per lap, aggregated across the 2018–2025 seasons. Early laps (red) and the primary pit window (orange) dominate; late-race changes (teal) are sparser.

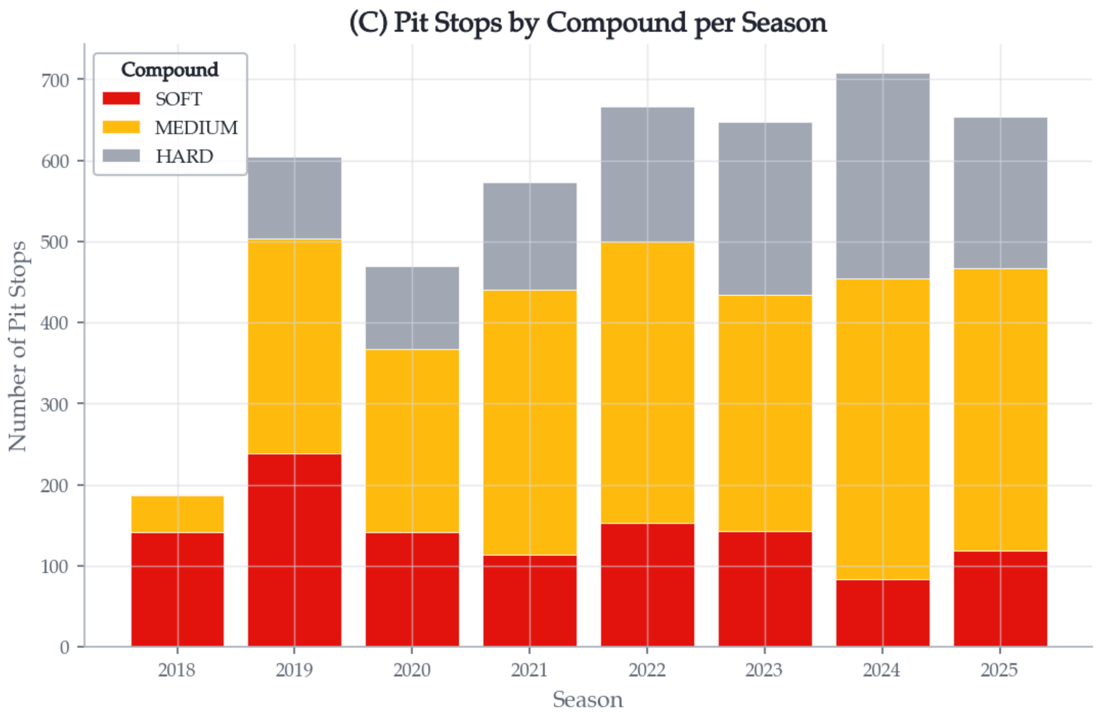


Figure 30. Pit-stop counts by tyre compound, aggregated per season (2018–2025). Medium has been the dominant pit compound since 2021; Soft usage peaked in 2019 before tapering, while Hard usage rose from a negligible share in 2018 to roughly a third of all stops by 2024–2025.

5.7 Descriptive analytics

Population-level descriptive statistics for the 2018–2025 subset are collected in Table 14. The primary pit window centres on lap ~ 28, with mean tyre life at pit ~ 15 laps.

Table 14. Population-level descriptive statistics, 2018–2025 seasons.

Analysis	Key finding
Lap time distribution	Mean of 89.6 seconds; bimodal peaks reflect compound and circuit differences
Pit lap distribution	Mean pit lap of 27.7; primary window concentrated between laps 20–35
Tyre life distribution	Mean of 15.1 laps; SOFT compounds pit significantly earlier than HARD compounds
Position change timing	Early laps (1–15) and mid-race laps (16–40) dominate net position changes
Class imbalance	4,818 pit events from 152,475 laps; 3.16% positive rate

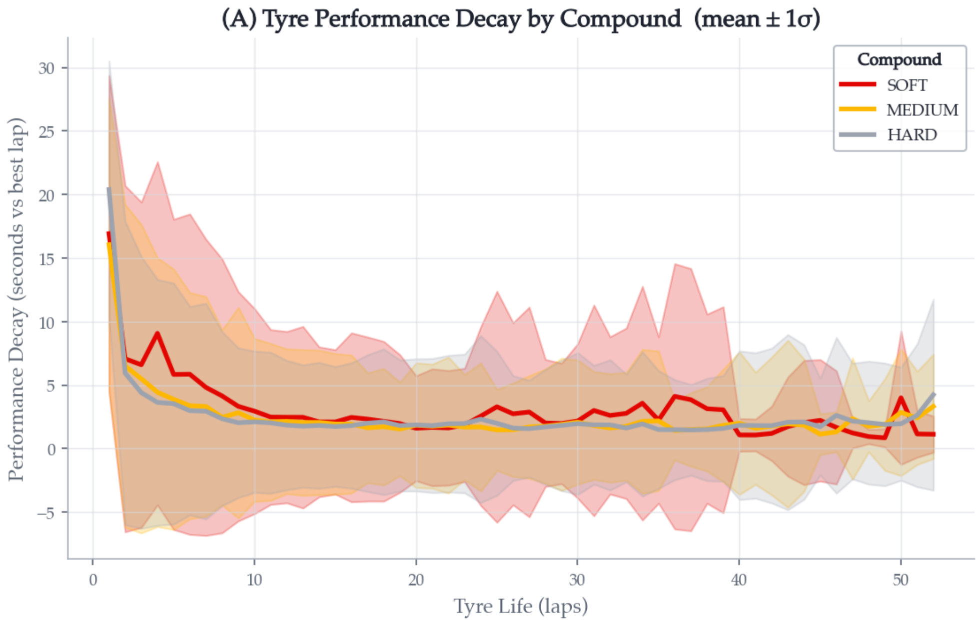


Figure 31. Compound-specific tyre degradation curves: mean performance loss (seconds vs. best lap) as a function of tyre age, with $\pm 1\sigma$ band. The Soft compound shows the steepest cliff.

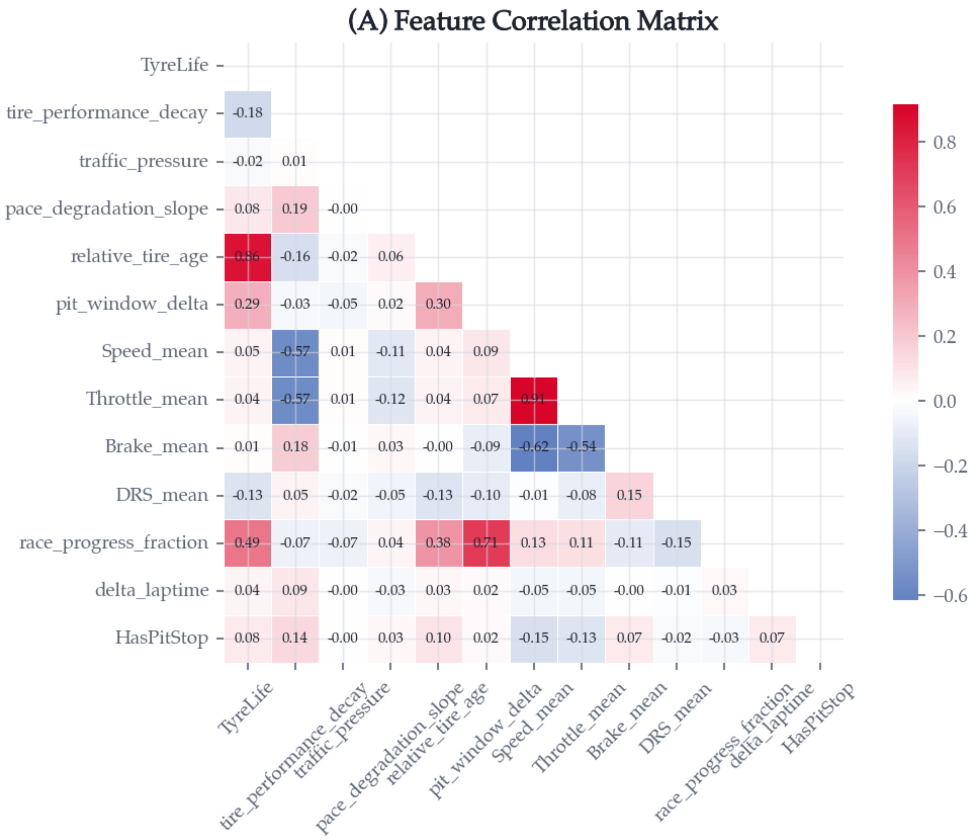


Figure 32. Pearson correlation between the engineered features (lower triangle). The strongest interactions are race-progress↔pit-window-Δ ($r = 0.72$) and race-progress↔tyre-life ($r = 0.49$); raw correlations with the pit-stop indicator are individually small, consistent with the BiLSTM's role being to learn the non-linear interaction structure rather than to amplify any single feature.

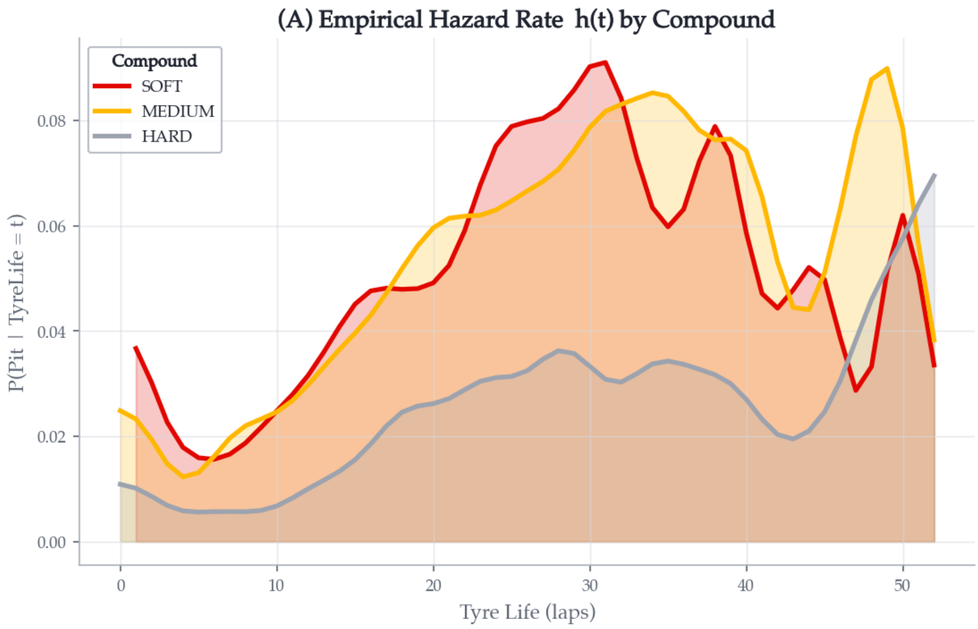


Figure 33. Empirical pit-hazard rate $h(t)$ as a function of tyre age, separated by compound. The Soft hazard rises earliest; the Hard much later.

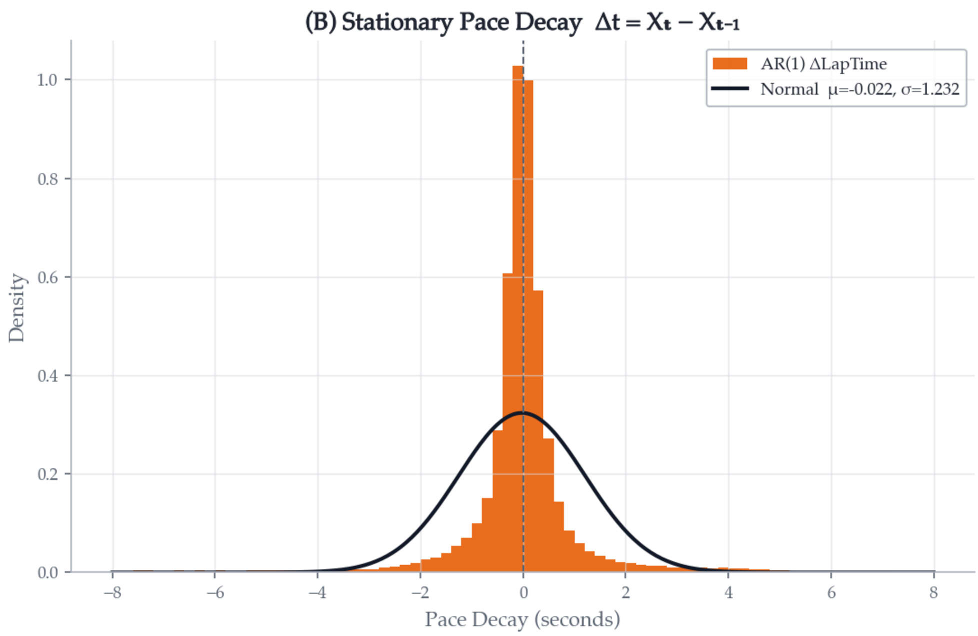


Figure 34. Stationary pace-decay $\Delta X_t = X_t - X_{t-1}$ after AR(1) filtering, with a Gaussian fit overlaid. The residuals are approximately symmetric around zero, validating the filter.

5.8 Discussion and future directions

The BiLSTM is operationally useful: on true pit laps in 2025 it assigns mean probability 84%, against a near-zero baseline on non-pit laps. The Cox model gives a complementary, interpretable hazard baseline that the deep model can be stacked against. Team profiles partition naturally into deterministic and opportunistic strategy cultures. The full pipeline — predicted pit probability, overtake classification, VSC agility score — supplies the building blocks for race-time strategic decision support.

A natural set of extensions follows.

- **Attention.** Replacing the BiLSTM stack with a small self-attention encoder would expose, lap-by-lap, which historical laps the model is conditioning on. Attention weights are far easier to debug than recurrent hidden states.
- **Multi-task heads.** A single backbone could simultaneously predict the binary pit indicator, the tyre-compound choice, and the residual stint length. Joint training tends to regularise each head.
- **Calibration.** Raw sigmoid outputs are not calibrated probabilities. Platt scaling or isotonic regression on the held-out season would yield probabilities that match empirical pit-stop frequencies, a prerequisite for use inside a strategy simulator.
- **Weather.** The single largest unmodelled covariate. Rain probability and track temperature reshape every part of the pit decision and would be the first feature to add in a richer pipeline.

6 Joint Discussion

The four studies above were performed independently and each used different statistical machinery. Read together, three cross-cutting observations stand out.

Machinery first, driver second. Sections 3–4 both demonstrate the dominance of the car. In Section 3, the constructor strength index alone explains more variance than any other feature in the linear regression, and grid position — itself largely a consequence of car capability — comes second. In Section 4 the teammate delta exists precisely to subtract the car off; once it is removed, the apparent home effect vanishes. A naive analysis that ignores the car will find a "home effect" for any driver who happens to have driven a competitive car at their home Grand Prix, and will find no effect for any driver who happened not to.

Era drift matters as much as model choice. Section 2 shows lap times tightening dramatically in the modern ground-effect era ($\sigma = 0.72$ s at Monza 2023 vs. 1.44 s in 2005). Section 4 requires era adjustment just to make the dependent variable coherent. Section 5 trains and tests across seasons explicitly because the pit-stop distribution shifts every year with tyre supply, sprint formats, and DRS rules. In each case, era is not a nuisance to be averaged over; it is part of the model's data-generating process.

The right baseline beats the right model. The pit-stop BiLSTM is more sophisticated than the lap-time MLE fits, but the operational uplift from each comes from the *baseline* the model is benchmarked against. A BiLSTM that beats a Cox proportional-hazards baseline by ΔAUC-PR points is a stronger claim than one that beats a constant predictor. Likewise, the home-advantage paired t-test would have been meaningless without first constructing the teammate delta, and the podium-decay analysis in Section 3 would be misleading without the empirical $P(\text{Win} \mid \text{Pole}) = 43\%$ anchor.

What we did not do. Three obvious omissions cut across the four studies. None of the analyses incorporate weather — wet-race outcomes are systematically different in distribution and would deserve a separate treatment. None of the analyses model strategic interactions across drivers (undercut races, blocking) explicitly. And none use FastF1-grade per-lap telemetry for the pre-2018 era, because it does not exist; the historical studies are therefore based on lower-resolution data than the modern pit-stop study.

7 Conclusions

Four studies, one substantive thread. The right choice of target variable does most of the methodological work in each.

- For **lap times**, the right target is the cleaned within-race distribution; the right model class is the right-skewed families (Gamma, Log-Normal), not the Normal.
- For **grid advantage**, the right target is the finishing position controlled for constructor strength; pole position remains the single dominant signal.
- For **home advantage**, the right target is the era-adjusted teammate delta, which removes both the car and the points-system confound; the aggregate home effect is then statistically zero, and only a small experience interaction survives.
- For **pit strategy**, the right target is the per-lap pit indicator on length-20 telemetry windows; a BiLSTM recovers an 84% mean probability on true pit laps.

The recurring message — in a sport where machinery and rules drift faster than driver skill — is that the modelling choice that matters most is the one made before any model is fit. The four analyses converge on this single point from very different statistical directions.

A Reproducibility

Software. All four pipelines run under Python 3.11/3.12. The shared scientific stack is `pandas`, `numpy`, `scipy`, `statsmodels`, `scikit-learn`, `matplotlib` and `seaborn`. The home-race study additionally uses `xgboost` and `shap`; the pit-stop study uses `torch`, `lifelines`, and (for raw data acquisition only) `fastf1`; the grid-position study uses `tensorflow` for its LSTM regressor. Random seeds are fixed to 42 throughout. For TensorFlow, `tf.config.experimental.enable_op_determinism()` forces deterministic kernels so that the held-out LSTM MAE is exactly reproducible at 2.87 positions.

Data. The Kaggle Ergast redistribution lives in `data/raw/`; a curated nationality-to-country mapping is shipped as `countries.csv` and used by the home-race study. The pit-stop subproject also requires per-lap telemetry CSVs from the FastF1 API (several gigabytes), which are not distributed in the repository because of their size; pre-trained BiLSTM weights are shipped under `outputs/pit_stop_strategy/models/` so that all figures can be regenerated without retraining.

Entry points. From the repository root,

```
python -m src.lap_time_distributions.lap_distributions
python -m src.starting_grid_advantage.main
python -m src.home_advantage.run_analysis
python -m src.pit_stop_strategy.visualize
```

each regenerate the artefacts under the corresponding `outputs/` subdirectory. Per-subproject `README.md` files document the full multi-stage commands where applicable.